

Final Year Project Report
XAI For Multimodal Fake News Detection on Social Media



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Dedication

In The Name of ALLAH, The Most Generous, The Most Merciful.

This is dedicated, first and foremost, to Almighty Allah, the origin of all knowledge, wisdom, and guidance, without Whose divine grace and benevolence no endeavor of consequences can be undertaken or completed.

To our honored parents and families, whose self-sacrifices, steadfast prayers and unwavering moral encouragement have constituted the very foundation of our academic activities. Their enduring commitment to our education and well-being has been the singular most significant driving force behind every achievement we have attained.

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This work stands as a testament to the collective investment of all those who believed in our potential and supported our academic journey.

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ABSTRACT

The rapid proliferation of social media has accelerated the dissemination of misinformation, in which fabricated or misleading claims are reinforced by accompanying imagery, exposes structural limitations of text-only detectors, which remain unable to recognize cross-modal inconsistency that are a hallmark of manipulated content, while deep classifiers that are able to detect both modalities typically behaves as opaque black-boxes, not well suited to accountable content moderation. this study addresses both limitations through an explainable yet accuracy-competitive detector. The proposed framework, HEMT-CLIP, fuses a fine-tuned RoBERTa-base text encoder with a CLIP ViT-B/16 vision encoder through a CLIP-similarity-gated cross-attention mechanism, in which the textual representation attends over the 196 image patch tokens and the attended representation is blended with the text under a parameter-free gate, $\text{fused} = \alpha \cdot \text{attended} + (1 - \alpha) \cdot \text{text}$, where α is the pre-computed CLIP cosine similarity. The model is trained on a stratified 17,149-sample subset of the Fakeddit corpus (70/15/15) using two-stage fine-tuning on a single T4 GPU. On the held-out test set ($n = 2,573$), HEMT-CLIP attains an F1 of 0.8393, an accuracy of 0.8313, and an AUC of 0.9122, surpassing both unimodal and concatenation baselines, while a controlled ablation isolates the similarity gate, rather than cross-attention alone, as the decisive component (+1.16 F1, +2.03 AUC). Notably, fake posts exhibit a higher mean text–image similarity (0.300) than real posts (0.253), inverting the conventional manipulation assumption, yet the gate succeeds as a learned soft blend; a three-level framework of cross-attention heatmaps, SHAP, and LIME further provides a conditional cross-method agreement analysis. The results establish that competitive accuracy and intrinsic, multi-granular explainability are jointly attainable under constrained academic resources.

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DEFINITIONS AND ACRONYM

<i>Acronyms</i>	<i>Meaning</i>	<i>Acronyms</i>	<i>Meaning</i>
ML	Machine Learning	NB	Naïve Bayes
DL	Deep Learning	DT	Decision Tree
NLP	Natural Language Processing	TTEC	Transformer-based Translation & Entire-Image Contrastive Framework
BERT	Bidirectional Encoder Representations Transformer	COOLANT	Cross-modal Learning for Multimodal Fake News Detection Alignment & Negative Training
mBERT	Multilingual BERT	MMFN	Multi-grained Multi-modal Fusion Network
Key-BERT	Keyword BERT	SAFE	Similarity-Aware Multi-modal Fake News Detection
RoBERTa	Robustly Optimized BERT Approach	Conv1D	1 D Convolutional Neural Network
LSTM	Long Short-Term Memory Networks	AUC	Area Under Curve
Bi-LSTM	Bi-Directional LSTM	F1	F1-Score
SHAP	Shapley Additive exPlanations	ROC	Receiving Operator Characteristics
LIME	Local Interpretable Model-agnostic Explanations	HEMT	Hybrid Explainable Multimodal Transformer
CLIP	Contrastive Language-Image Pre-training	LLM	Large Language Model
XLM-R	Cross-Lingual Language Model with RoBERTa	XAI	Explainable Artificial Intelligence
SLMs	Small Language Models	dEFEND	Explainable Fake News Detection
Res-Net	Residual Network	Ex-Fake	Explainable Fake News Detection Framework
BLIP	Bootstrapping Language-Image Pre-training	TELLER	Textual Evidence-based Learning for Fake News Recognition
XL-NET	eXtreme Long-Net	GACoNF	Graph-Based Attention Convolutional Network for fake News
Grad-CAM	Gradient-weighted Class Activation Mapping	Cro-Me	Cross-Modal Tri-Transformer and Metric Learning
CNN-LSTM	Convolutional Neural Network LSTM	SR-CIBN	Semantic Relationship-based Consistency and Inconsistency Balancing Network
LDA	Latent Dirichlet Allocation	SGAMF	Sparse Gated Attention-Based Multimodal Fusion Method for Fake News Detection
Distil-BERT	Distilled BERT	XFND	Explainable Fake News Detection
RNN	Recurrent Neural Network	BC-FND	Bi-directional Cross-Modal Fake News Detection
GraphSAGE	Graph Sample and aggregatE	CMFNThinker	Cross-Modal Fake News Thinker
GAT	Graph Attention Network	MEFaND	Multi-evidence Fusion for Fake News Detection
ALBERT	A Lite BERT	CoDE-DiDA	Contextual Deep Ensemble for Disinformation Detection and Analysis
T5	Text-to-Text Transfer Transformer	RAFAH	Relation-Aware Fake News Detection with Hierarchical Attention Propagation Reasoning Units
GRU	Gated Recurrent Unit	PRUs	Deep Fake News with Semantic Correlation Neural Classifier
Bi-GRU	Bi-Directional GRU	DFN-SCNC	Dual GCN-based GNN with Gated Recurrent Unit
Gated-GNN	Gated Graph Neural Network	DGBM	Fake Content Detection using CLIP
GCN	Graph Convolutional Network	FACT-CLIP	Feature-level Multimodal Correlation
VAE-GCN	Variational Auto-Encoder GCN	FMC	Feature Cross-Modal Matching
RvNN	Recursive Neural Network	FCM	Semantic Web and Reasoning-Oriented Graph Framework
EANN	Event Adversarial Neural Network	SWAROG	Word Embedding Over Linguistics Features
DANN	Domain-Adversarial Neural Network	WELFAKE	Multi-Layer Perception
BDANN	Bidirectional Domain-Adversarial Neural Network	MLP	Large-Language Model Meta-AI 3-70B parameters
S-BERT	Sentence BERT	LLaMA3-70B	Precision, Recall
SVMs	Support Vector Machines	P, R	
GRN	Generative Round Network		
HAN	Hierarchical Attention Network		
XGBoost	Extreme Gradient Boosting		
VGG19	Visual Geometry Group 19-layer Network		
GloVe	Global Vectors for Word Representation		
GPT	Generative Pre-trained Transformer		
RF	Random Forest		
LR	Logistics Regression		

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1 INTRODUCTION

The drastic increase in social media platforms has transformed the manner in which information is spread but has also increased the propagation of misinformation and fake news. The unfiltered and high-speed nature of social media enables the dissemination of misinformation to millions before fact checkers can act and shape the opinions of the masses on politics, healthcare decisions. Conventional machine learning and deep learning algorithms have been useful in detecting fake news, and their opaque nature does not allow users to determine the logic behind their predictions. Such lack of transparency makes users less trustful and makes AI-based systems of misinformation detection less acceptable.

The project suggests the design and development of an Explainable Artificial Intelligence XAI system to detect fake news on social media. The goal of the system goes beyond classifying the news as either “real” or “fake” and clarifying the reasons behind the classification citing important features of the language and context that informed the choice. This system will fill the gap between the performance and interpretability of models by incorporating the use of transformer-based models like RoBERTa with explainability frameworks like SHAP and LIME. The proposed system's explainable design will make things more transparent and increase the trust of the users as well as promote responsible AI practices in social media analytics.

1.1 Problem Overview

The current models of fake news detection are accurate but do not explain their choice of classification. They are based on opaque deep learning models that do not offer interpretive reasons behind their outputs and therefore, the predictions cannot be understood or trusted by the users and policymakers. Moreover, the majority of these models are only trained on English data and they fail to generalize across languages especially in low-resource languages such as Urdu. Also, there is the issue of disregarding contextual factors like writing style, tone of emotion, credibility of the source etc. This brings a pressing necessity to systems that can merge linguistic intelligence with contextual knowledge and explicability to allow human insights to interpret the findings of AI. Therefore, the suggested system will apply the Explainable AI techniques to establish transparency, trust, and accountability in automated fake news detection.

1.2 Research Questions

The proposed research will help to answer the following main research questions:

1. What can be done to enhance transparency and trust in the fake news detection systems using Explainable AI techniques?
2. What are the strongest linguistic, contextual and semantic characteristics that drive the detection of misinformation on social media?
3. How can cross-modal similarity be incorporated into a text-image fusion mechanism, and how can intrinsic attention together with post-hoc methods such as SHAP and LIME be used to interpret the resulting predictions?
4. Which evaluation metrics best capture both performance and quality of explanation of the system produces?

1.3 Research Objectives

The primary aims of research are to create an Explainable AI model that will be capable of identifying fake news in an accurate and understandable manner. In particular, the following are the objectives:

1. To develop a multimodal fake news detection model that fuses a fine-tuned RoBERTa text encoder with CLIP ViT-B/16 image encoder through a CLIP-similarity-gated cross attention mechanism.
2. To implement three-level explainability framework combining intrinsic cross-attention heatmaps, SHAP and LIME in order to visualize and interpret model decisions across both modalities.
3. To determine and specify linguistic, visual and cross-model cues that drive fake news classification and to quantify the contribution of each modality through a controlled ablation study.

1.4 Scope

The suggested study is devoted to the detection and explanation of multimodal misinformation on social media using the Reddit-sourced Fakeddit corpus as its benchmark. It pairs a fine-tuned RoBERTa text encoder with a CLIP ViT-b/16 image encoder through a CLIP similarity-gated cross-attention fusion mechanism, and incorporates three-level explainability framework in generating readable results and pictorial explanations for every classification.

The project is restricted to fixed single image–text pairs and does not consider video-based disinformation such as deepfakes, multi-image posts, or pixel-level image-forgery forensics. Classification is binary (real versus fake), so finer categories such as satire, misleading context, and partially accurate claims are out of scope. The system is evaluated under controlled, offline conditions rather than on live, real-time social-media streams, and it operates on English-language content only. Nevertheless, the model is designed to be scalable for incorporation into live systems in future.

1.5 Methodology

The research will follow a mixed method approach combining machine learning model development and explainability analysis. Data will rely solely on publicly available Fakeddit dataset, a large scale multimodal for fake news detection. Social media data will also be scraped using APIs and annotated using verified fact-checking sites.

Then the preprocessing stage, the preprocessing stage parallel-downloads and resizes images to 224×224 with CLIP normalization, tokenizes post titles with the RoBERTa tokenizer (128-token cap), precomputes a per-sample CLIP text–image similarity scalar, and packs the aligned text, images, similarity values, and labels into HDF5 files with a stratified 70/15/15 train/validation/test split. The multimodal model, a fine-tuned RoBERTa text encoder fused with a CLIP ViT-B/16 image encoder via similarity-gated cross-attention, is then trained to classify posts as either genuine or fabricated.

The project aims to clarify by using SHAP alongside LIME i.e., tools that reveal which factors matter most, even down to individual words. These insights will be shown visually, spotlighting what in the language led to each decision. We’ll measure how well the system performs using numbers like accuracy, F1-score, and also ways to measure clarity.

1.6 Significance/ Potential Applications

The project holds both for learning and real life. It builds trustworthy AI that fights false information, joining a global push for sensible rules around AI. Because it explains its reasonings clearly, people, including reporters and those making laws, can grasp how the AI arrives at its conclusions.

2 BACKGROUND

The proliferation of fake news on social media platforms has emerged as one of the defining challenges of the digital era. Unlike traditional media, online platforms allow content to be disseminated instantly, at negligible cost that can bypass editorial gatekeeping, making that fabricated or misleading information virtually indistinguishable from genuine reporting when it reaches the viewer/reader. The Reuters Institute and others have revealed that false information travels more rapidly and widely than corrections, creating asymmetric information environments that have real-world effects ranging from public health to electoral integrity, financial markets, or social cohesion. The economic impact of misinformation has been extrapolated to costing billions of dollars each year when downstream effects on health choices, investment behavior, and institutional trust are added up.

This is further complicated by the multi-modal character of today's misinformation. Today, artificial intelligence has been used to match text with images in fake-news posts on platforms like Reddit, Twitter (X) and Facebook, taking advantage of what humans tend to believe more: a visual element that accompanies text. The presence of multimodal fake news complicates this problem because the cross-modal inconsistency is often used: falsified articles are often accompanied by genuine pictures and falsified text by emotionally appealing images, or authentic pictures are used to add some counterfeit believability to falsified text. Detection methods based on unimodal (image only or text only) data are structurally ill adapted to the problem, because they cannot represent the semantic relationship between modalities, which forms the main focus of manipulation. To deal with such content, systems are required to reason over content from both modalities, not separately over text or image classifiers.

This chapter provides conceptual and technical background: (1) establishing the social-media misinformation landscape; (2) defining fake news and the multimodal case of the misinformation problem; (3) the generational evolution of detection architectures; (4) the cognitive psychology of the misinformation problem and susceptibility; (5) the XAI imperative; and (6) the reference dataset and bounding constraints.

2.1 The Social Media Misinformation Landscape

In recent times, the internet has influenced how individuals engage and communicate due to its ease of accessibility, low cost, and rapid transmission of data. As a result, many individuals now prefer using social media and online portals to read and search for news of interest rather than traditional newspapers. The increasing speed of social media development, Facebook, X (and formerly Twitter), Instagram, Tik Tok have completely changed the way information is produced and consumed. The algorithmically selected feeds have become the primary news source of billions of users and do little to verify the content. The subsequent outcomes are largely reported: fake news spreads faster than actual news and reaches more people compared to real news, mainly because of the human element of sharing as opposed to the use of automated bots. Such prominent high-profile examples are the misinformation of the 2016 U.S. presidential election and the COVID-19 infodemic, where health misinformation was directly responsible for vaccine hesitancy and harm that could have been mitigated. A recent study by MIT researchers of 126,000 Twitter cascades determined that false stories spread, reached 1,500 recipients about six times more rapidly than true stories, and that people shared the false stories rather than bots, and that the process was “affect-driven” a novel, arousing, moral message spreads before a deliberative assessment takes over. Most importantly, the fake news in the present-day world no longer exists as only a text but is combined using images, video, and emotionally evocative framing into multimodal working constructs, which are more convincing and much harder to notice than text-based fake news.

2.2 Defining Fake news and cross-modal manipulation

Fake news has an operationally defined meaning as fabricated or misleading content that is intended to deceive the reader or to influence the opinion of the masses and does not include satire, parody or a legitimate journalistic error. The multimodal architecture is motivated by the manipulation pattern of Cross-modal inconsistency. In practice there are four characteristic forms:

- **Real image, fabricated caption:** a real photograph is used with a false headline, deception is completely in the headline.
- **Out-of-context image, factual claim:** manipulating or providing an irrelevant picture gives emotional credibility to a factual claim.
- **Semantically aligned fake content:** Text and image are made up but similar, with high CLIP similarity; the most common case in Fakeddit, as fake posts have higher mean α (0.300) than real posts (0.253).
- **Manipulated framing:** authentic content is presented in a selective framing or with emotionally charged language which distorts the meaning of the content.

These text-only detectors are structurally blind to all four categories as they are based on the relationship between text and image, not within the text or the image. This is not a performance issue, but an architectural one, as described above: the main motivation for the cross-attention fusion design.

2.3 Evolution of fake-news detection Architectures

Automated detection has progressed through four identifiable generations. Table 2.1 summarises the progression:

Generation	Models	Key Strength	Key limitation
Gen1-Recurrent	LSTM, Bi-LSTM, GRU	Sequential pattern capture	No long-range context, text-only
Gen 2-Transformer	BERT, Roberta, XLNet	Deep contextual representations	Text-only, black-box decisions
Gen 3-Multimodal fusion	CLIP, FND-CLIP, FACT-CLIP	Cross-modal evidence reasoning	Often no intrinsic explainability
Gen 4-Explainable Multimodal	HEMT-CLIP (this work)	Detection + three-level XAI integrated	Single seed, English, Reddit only

First generation recurrent models (LSTM, Bi-LSTM) learned sequential linguistic patterns, but suffered from vanishing gradients on long texts and were purely text based. Large performance gains were achieved with second generation transformer models (BERT, RoBERTa, XLNet) which are opaque and purely text-based. For its powerful pretraining on noisy web data and stronger training procedure (no NSP, larger batches), RoBERTa-base is utilized in HEMT-CLIP. Third generation multimodal fusion systems combine text and image information, consisting of early fusion (concatenation, fast to compute but does not model fine-grained cross-modal relations) and of cross-attention fusion (text queries image patch tokens that can model cross-modal relations and localization and intrinsically image heatmaps). With 400 million image-text pairs, CLIP is easily trained to enable cross-attention fusion even at the scale of academic research, by learning to align across a shared semantic space that serves as an alignment signal, while a ViT-B/16 vision encoder produces 196 spatial patch tokens for localized attention. The closest conceptually, HEMT-Fake [35] is a multimodal, multilingual transformer with token, sentence and modality-level

explainability features that use SHAP+LIME. HEMT-CLIP differs by using CLIP similarity as a parameter-free fusion gate and targets the English Fakeddit benchmark.

2.4 Cognitive psychology of misinformation.

According to Pennycook and Rand's dual process theory, misinformation sharing is largely automatic and motivated by affect (emotion) and less by partisanship or ideology: Affect is the primary factor for people pausing to check the accuracy before they share, and people who do pause to check are significantly less likely to share false content, across partisanship. The findings show that the sharing rate of fake news rises about 20% on average when a single moral-emotional word appears in the post. The results demonstrate that moral-emotional words are indeed a statistically predictive element of fake news, since they lead to an average increase in sharing rate of 20% when present in a single post. The implications for HEMT-CLIP are twofold: First, since RoBERTa has been pre-trained on emotionally diverse web text, it is able to score well on determining whether the framing is emotional or factual; Second, the vocabulary-bias error discussed in Chapter 6, the text model's tendency to classify real texts about Kristallnacht as fake is exactly what dual-process theory predicts. The image channel that presents the evidence of the historical context corrects this failure.

2.5 Explainability related to Fake-News Detection

A major obstacle to the deployment of deep-learning classifiers in the domains where accountability is an issue is that of black-box classifiers. Content moderation impacts individuals and public discourse, and this is becoming the case with ever-expanding frameworks like the EU AI Act, which demand explanations. XAI approaches can be divided into three categories: intrinsic (e.g. cross-attention heatmaps) are a form of explanation that is a by-product of the architecture, and do not require access to the model internals; perturbation-based (e.g. SHAP, LIME) are model-agnostic and require no access to the model internals; and gradient-based (e.g. Grad-CAM, Integrated Gradients) require access to the model internals.

SHAP (partition / Owen-value algorithm) estimates Shapley values by masking token subsets, meaning that it takes the cooperative game theory axioms (efficiency, symmetry, linearity) into account, but operates at BPE sub-word granularity allowing to break up words. LIME provides a locally weighted linear surrogate of the word deletion perturbations, which it operates on at the whole-word level, which are less mathematically rigorous but more readable. SHAP proved to be more stable and LIME was faster (less than 0.5 s per sample, vs 1–5 s per sample) in a systematic comparison. On the same 30 stratified test samples, HEMT-CLIP is used to evaluate both in order to get a concurrence that will help build confidence in the attribution and to get a divergence, which is itself informative. XAI for fake-news detection is not widely used (only 11 models are genuinely explainable, from hundreds of reviewed models, 2018-2022, 7 of which exclusively used attention weights). HEMT-CLIP offers all three levels (intrinsic heatmap + SHAP + LIME) simultaneously.

2.6 Reference Dataset and Bounding Constraints

The largest publicly available truly multimodal fake-news corpus, which was constructed from Reddit posts marked with sourcing subreddit, is used by the HEMT-CLIP. It is selected for its size, as it comes from existing social media, for its 2-way labels that are clean, and for its being used in similar previous research (Tri-FusionDet [10], BC-FND [11]). The processed corpus contains 17,149 samples (8,716 real / 8,433 fake; 50.83/49.17%) split 70/15/15 into 12,003 training, 2,573 validation, and 2,573 test samples. Key statistics: The median length of the titles is 10 RoBERTa tokens (which indicates that image is the predominant information channel in titles); the mean CLIP α is 0.276 (std 0.054, range [0.086, 0.490]); the mean α of the fake titles is 0.300 (std 0.054, range

[0.268, 0.490]), which is counter-intuitive as discussed in Chapter 4, and the mean α of the real titles is 0.253 (std 0.054, range [0.167, 0.423]).

All the design decisions are subject to four bounding constraints. Language: English only (due to maturity of the datasets and tool chain). Platform: Reddit posts on social-media only. It is worth noting that the transformer-based models were selected over classical ML models for the task of contextual understanding, with RoBERTa-base and CLIP ViT-B/16 being deployed over the bigger models GPT-4 and LLaMA, due to the cost constraint of having only one free tier T4 GPU. Ethics: The algorithmic bias and the risk of censorship are given first-class treatment, as are sub-types of partial-truth and of satire and misleading context; the binary classification scheme is explicitly recognized as a simplification.

3 LITERATURE REVIEW

This research article [1] was the first to combine social context with explainable misinformation detection based on the DEFEND framework in which the end user comments are used as an explanation of why a news article is viewed as fake. The authors appreciate that crowdsourced verification clues in user interaction tend to exist, thus they have created a hierarchical co-attention network that is capable of modelling both news content (through a HAN) and user remarks. Such architecture is a dynamically updating semantic connection between particular sentences in a news item and the comments, basically showing how the wisdom of the crowd can be used to interpret the content. The experimental findings of PolitiFact and GossipCop indicated that DEFEND not only achieved more than 5% higher F1 -score than the state-of-the-art baselines but also effectively retrieved the top-K most explanatory comments and did so with high accuracy. The framework, however, is based on the quality of user comments and can only be used in the early news distribution when there is a low participation (the cold-start problem). Furthermore, when analyzing the issue based on text only, one misses the fact that the visual manipulation is becoming increasingly prominent in the current multimodal disinformation campaigns. This highlights why it is important that the future models can be extended to the realm of visual to make them truly explainable.

In [2], recent research has carried out an extensive study of the multimodal fake news detection methods, proposing a multi-image framework, which, unlike current methods, can draw visual characteristics of only one image and disregards the abundance of information which may be presented in the articles with multiple images. The research underlined that earlier multimodal systems, even though they were text and visual in nature, did not capitalize on the sequential and cumulative visual information available in many images that are embedded in an online news story. Not only the suggested architecture makes use of the BERT to retrieve contextual and semantic textual features and VGG-16 to retrieve visual features of multiple images processing by LSTM layers to capture the temporal sequences, but also proposes the idea of text-image similarity as a semantic feature that is computed, using the cosine similarity of the title embedding and image tag embedding. Test results on the GossipCop dataset were found to be superior to the state-of-the-art algorithms with statistically significant 5.39% higher F1-score than SpotFake and 4.19% higher F1-score than BERT baseline. Nonetheless, although the multi-image approach evolves visual feature extraction by processing multiple images sequentially up to three images and attention-based fusion, it is domain-specific to celebrity and entertainment news on GossipCop, utilizes under-sampling to overcome the issue of class imbalance that can cut useful training data and fails to pivot on the challenge of identifying advanced image manipulations and deepfakes produced by current AI-based technology, a critical challenge when countering the changing multimodal misinformation environment where participants deliberately construct misleading text-image fuses. This underscores the need to have stronger frameworks that integrate superior image forensics, capability to apply cross-domain generalization and sophisticated mechanisms in identifying subtle inconsistencies between textual assertion and visual evidence in various news areas and social media services.

This study [3] addressed the issue of effective measurement of cross-modal correlation in the context of misinformation detection by introducing a framework named FND-CLIP, which is based on the Contrastive Language-Image Pretraining (CLIP) concept. There is a combination of ResNet and BERT encoders with pairwise CLIP encoders in their study to obtain deep representations and presented a novel procedure to weigh multimodal representations with dynamically weights depending on the similarity scores that CLIP gives. And also enhanced this method by introducing an attention-like feature by combining text and image features on-the-fly, and fused feature weighted by relevance to each particular news item. The experiments indicated that FND -CLIP

had better results in accuracy by 0.7, 6.8 and 1.3 percentage points on the Weibo, PolitiFact and GossipCop datasets, respectively, as compared to the leading methods, and was effective to extract valuable cross-model features. Although the zero-shot capability provided by CLIP is quite strong, the investigation is also limited per se by the domain knowledge of the pre-trained model, and the use of standard benchmark datasets, which can limit the ability to expand its use to larger or more counter-intuitive misinformation settings. In turn, future studies must aim at enhancing large-scale multimodal models to more specialized situations in news-verification.

The Stylometric and Semantic Similarity-oriented Multimodal Fake News Detection (SSM) model proposed by [4] fills the gaps in existing methodological approaches that insufficiently consider the visual content through a set of generic pre-trained convolutional neural network (CNN) feature maps or consider fake-news detection as a traditional natural language processing (NLP) issue. The paper establishes that earlier multimodal systems, such as SAFE, rely on pre-trained image-to-sequence networks that use objective declarative signals instead of the emotive signals needed to enhance correct context-text interaction, and they do not effectively model the context-text interaction. SSM comprises five interdependent modules, namely: a Hyperbolic Hierarchical Attention Network (Hype-HAN) to extract stylometric textual features; a Text Rank-based summarization module with cosine-similarity matching between headlines and generated summaries, a token overlap of about 12 -percent at a similarity threshold of 0.75; a semantic image feature extraction module via EfficientNetB7, with textual summaries as the evaluation task, using a joint loss function weighted at a =0.7 and a =0.3 The classification accuracy of the FNDD data, TI- CNN and FNSDS were 96.74% and 96.50% and 96.90% respectively with ROC-AUC values of between 95-98, significantly higher than the accuracy of 94.74 95.90 of contemporary ensemble hierarchical attention network (HAN) methods. However, with these encouraging findings, the method has high computational costs because of its multiple processing streams, is tuned manually, is based on particular pre-trained architectures, and is only tested on English-language corpora-problematic areas that require additional optimization to run in real-time and cross-lingual generalization.

The research reported in [5] addressed the increasing complexity of misinformation generated by AI technologies by proposing a novel explainable multimodal fake post detection method. Recognizing that existing unimodal approaches fail to capture the subtle interplay between text and images in sophisticated forgeries, the authors constructed a hybrid dataset mixing human-created and AI-generated fake content. Their proposed framework incorporates image captions as auxiliary features and utilizes a multi-head cross-attention mechanism to evaluate micro-level cues spanning different modalities. Experimental results demonstrated that this approach improved the F1 score by approximately 10.2 points compared to baseline models. Furthermore, the integration of Attention maps for text and Grad-CAM for images allowed the model to provide visual and textual evidence for its decisions, thereby enhancing interpretability. However, the study's reliance on a specific, newly constructed dataset and its primary dissemination via technical reports suggest a need for broader validation against standard benchmarks to fully establish its generalizability and robustness in diverse real-world scenarios.

The work reported in [6] have created a unified model that incorporates the DAN, alongside the LIME-based explainability to identify misinformation of any kind across various social media platforms and provide clear explanations of the decisions made. The authors found severe weaknesses in the current methods, especially platform-specific detectors that need to be trained individually per network and black-box models that dismiss understandable reasons. Their methodology uses unsupervised domain adaptation where News and Instagram data are used as source domain to learn domain-invariant features and Twitter, Reddit and YouTube are used as target domain. The architecture combines Conv1D layers, LSTM networks as well as a gradient reversal layer which maximizes domain confusion and minimizes the error of classification, thus allowing cross-platform generalization with no target domain annotation. CoAID experiments and

new MiSoVac dataset (190,000 COVID-19 vaccine postings) experiments showed the superiority of DANN, with 40%F1-score and 3% accuracy improvement and 9% AUC increments. The implementation of LIME gives instance-level explanations based on locally weighted linear approximations with Random Forest surrogates, visualizing features that are influential with visualizations probability-weighted. Nevertheless, the research recognizes the instability of LIME because of random perturbation sampling and insufficient validation using two datasets. The strategy has economic benefits as it removes the platform-specific training and annotation costs, and practical implementation is needed to overcome the computational overhead and to generalize the validation to healthcare settings.

In [7], addressed the issue of “black box” nature of misinformation detection by suggesting ExFake, a hybrid framework that integrates content analysis with social context and XAI. The authors understand that automated detection is not sufficient to transform user behavior or change it, thus developing a system that does not only identify news veracity but also assesses the credibility of the sharing user and cross-reference content with named entities that can be trusted and accredited fact-checking organizations. One important innovation of ExFake is the in-built XAI assistant that supports interpretable (rationales) behind the decisions it gives, and it is designed to subtly encourage users to think critically instead of filtering content. Real-world dataset experimental results showed that ExFake is substantially more effective than the baseline approaches in harnessing the social history of users and knowledge graphs. Nonetheless, the reliance of the system on continuous information flows of third-party fact-checkers and previous user information indicates that it might be limited in real-time scalability and the ability to apply it to new accounts (the cold-start problem), which will require more independent, zero-shot explanatory processes in future versions.

The research described in [8] addresses the urgency to establish the credibility of automated misinformation detection by developing TELLER, a dual-system model that was to be explainable, generalizable, and controllable concurrently. The authors also inspired the dual-process cognitive theory, instead of a black box nature of traditional DL, to design a two-stage architecture. The initial step is the Cognition System that uses human knowledge to create rational Yes/No question templates (predicates) that are answered with LLMs to produce explainable logic atoms. The second system, the Decision System is a differentiable neural-symbolic model that uses these atoms to compile them with domain-invariant logic rules to more or less infer the truth of news. Experiments on four benchmarks proved that TELLER is not only as accurate as the best state-of-the-art methods, it also has comprehensible paths of reasoning capable of controllability and generalizable. Nevertheless, the reliance of the framework on the use of LLAMs as the first step of thought presents a liability of being dependent on the factual accuracy of the underlying model, and the handwriting of logic predicates can present scaling issues to handle new and fast-changing types of disinformation that do not fit in the existing logic templates. This hints to the direction of future research of automating predicate generation and improving the self-correction capabilities of the neuro-symbolic reasoning engine.

The work presented in [9] responded to the critical need for robust and interpretable misinformation detection by evaluating a diverse suite of ML and DL architectures enhanced with FastText embeddings. Their comprehensive study, conducted across three major datasets. WELFake, FakeNewsNet, and FakeNewsPrediction, demonstrated that a hybrid CNN-LSTM model significantly outperformed traditional ML algorithms and even rivaled sophisticated transformer-based models (BERT, XLNet, RoBERTa) in specific configurations, achieving near-perfect accuracies of 0.99, 0.97, and 0.99 respectively. Beyond mere classification, the authors addressed the “black box” issue by incorporating XAI techniques, specifically LIME and LDA, to visualize influential keywords and thematic topics driving the model's decisions. However, while the study establishes a strong benchmark for text-based detection, its focus remains largely unimodal, potentially missing the nuanced, cross-modal inconsistencies found in modern multimedia

disinformation campaigns, thus suggesting a need for future work to extend these hybrid architectures and explainability frameworks to image and video domains.

The study of multimodal fake news detection methods by [10] is a thorough one: Tri-FusionDet is one of the schemes that fill the significant gap in current approaches by combining textual, visual, and social engagement characteristics by using a late fusion approach. Their research pointed out the weaknesses of earlier multimodal approaches that mostly concentrated on text-image pairs and disregarded other useful social engagement indicators like comments, likes, and ratios of upvotes. The suggested late-fusion architecture does not only allow processing each of the modalities independently, but also allows one to effectively incorporate them at the decision-making level, which enhances the accuracy of the detection but does not present the dimensionality issues that early fusion strategies do. Their testing on large scale Fakeddit data showed results that were better than the state-of-the-art methods with an accuracy of 94 and a 6% improvement over the highest performance baseline model. Nevertheless, whereas Tri-FusionDet builds upon multimodal fusion by its tri-modal fusion and late fusion method, it mainly depends on pre-trained models (Distil-BERT and ResNet18) without the need to investigate domain-specific feature engineering and sophisticated attention mechanisms that are important to address complex inter-modal interactions, which the ever-growing fake news generated by GenAI technologies are becoming. This underscores the role of additional adaptive fusion methods utilizing dynamic feature selection process and explainability methods to improve detection and model explainability in changing misinformation settings.

The research reported in [11] performed extensive research of the multimodal fake news detection approaches, presenting a set known as BC-FND, which combines the textual and visual data into one embedding space and manages to effectively integrate them. The work highlighted the shortcomings of current methods that use disparate feature extractors on both text and images and it is hard to match extracted features to shared semantic spaces as well as use sophisticated fusion techniques that create redundancy and noise. Not only does the hierarchical bilinear fusion strategy allow the fine-grained complementarity of BERT-extracted textual features with CLIP-aligned multimodal features but it also includes a consistency loss function to train dissimilarity patterns between fake news text-image pairs and similarity patterns in real news. And were tested on two publicly available datasets and the newly introduced Weibo23 dataset with better performance over state-of-the-art methods, and improvements were particularly pronounced in the detection of fake news that included document images, e.g., chat screenshots. Nevertheless, BC-FND has serious performance degradation when using the temporally-divided data, where the content distribution varies with time, and fails to perform optimally in forward-looking prediction tasks, which would be important in real world application in dynamic social media settings where patterns of fake news are changing on-the-fly. This brings to the fore the necessity of more adaptive schemes to include time-invariant feature learning schemes and external knowledge checks to ensure that strong detection performance is achieved when there is shift in the content between times and uncovering a new misinformation sub-strategy.

The work presented in [12] thoroughly examined multimodal early fake news detection approaches and presented MEFaND, a system that can detect fake news in constrained timeframes with limited information about the propagation of one social media post. In their research, they highlighted why the current methods are incomplete to either only use one (textual content or propagation patterns) or need a large amount of propagation data between posts to work, which is unfeasible when news propagates at the fastest speeds. The given multimodule architecture does not only use GNNs (GraphSAGE and GAT) to learn edge-enhanced and node-enhanced propagation graphs based on temporal, relational, sentimental information, but with the help of BERT, it generates contextualized representations of short-length news. The analysis on PolitiFact and GossipCop datasets proved better performance than state-of-the-art techniques with the F1-score of 99% and 96% respectively using a five-hour detection deadline. Nevertheless, although MEFaND promotes

early detection based on the overall analysis of user characteristics and the temporal-structural pattern of propagation, it excludes visual modalities but not the multilingual and multiplatform nature of social media streams, even though these features are essential in the application of the system in the rapidly changing environment of Gen AI-based fake news. This implies the necessity of more generalized frameworks that integrate visual content analysis, domain specific adaptation schemes, and greater computational efficiency to support the global, real-time dynamic nature of fake news communication across the various social media environments.

The work presented in [13] thoroughly analyzed the fake news detection based on transformers, and they proposed RAFAH, a model that integrates the fine-tuned RoBERTa architecture along with hybrid features extraction to overcome the shortcomings of the current content-based detection frameworks. The model used a multi-dimensional feature engineering strategy that derived linguistic, psycholinguistic, emotion, and readability features out of the news content and they were concatenated with the news titles then fed through the fine-tuned RoBERTa with attention mechanisms. The framework showed better performance in two benchmark datasets with 89% acc. on BuzzFeed and 99% on ISOT (Bi-LSTM, LSTM, CNN-LSTM and traditional ML methods by 10% and 1%). The study highlighted that, transformer models are better at addressing complex semantic patterns in comparison to sequential models such as Bi-GRU which were only able to reach an accuracy of 65% on BuzzFeed since they do not support integration of multi-features and cannot detect subtle linguistic patterns. Nevertheless, although RAFAH proceeds to detect fake news using a holistic feature engineering and contextualized embeddings, the model only considers text-based features, but not multimodal (images or videos) ones, and does not examine the patterns of social propagation and ecosystem dynamics that define the spread of misinformation. This points to the necessity of future solutions that combine multimodal properties, understand social patterns of engagement, scale to multilingual data, and deploy applications in the real world so as to deliver the complete fake news detection capabilities on social media platforms and across the variety of content and distribution channels.

Prior study [14] have carried out an in-depth study of explainable fake news detection, providing a pyramidal co-attention model, which fills the essential gap in the knowledge of why this or that news item is categorized as fake. Their article used PRUs with HANs and sentence-comment co-attention to learn together on the FakeNewsNet dataset that consists of PolitiFact and GossipCop sources. The framework performed better, being top by 5.56% points in terms of F1 score and also finding explanatory power as the top-ranking identified the check-worthy sentences, and the influential user comment. The study has stressed that the PRUs address the generalizability limitations of conventional RNNs and LSTMs by using pyramidal transformation and grouped linear transformation to facilitate the extraction of the features more efficiently on multiple scales. The explainability of the framework was validated through human assessment with Amazon Mechanical Turk that showed 16.4% and 22.1% increases in NDCG and precision of top-k explainable comments respectively, over baselines with 122 out of 150 workers favoring the rankings of the comments proposed by the model. Nevertheless, although the framework enhances the accuracy of detection and interpretability by using co-attention to form mutual relationships between user responses and news sentences, it uses Claim-Buster to verify the ground-truth sentence and fails to integrate external fact-checking knowledge bases during training. This underscores the importance of future strategies that combine sources of authority fact-checking directly into the learning process and evaluation of commenter credibility to promote the detection reliability and the quality of explanation of various misinformation scenario.

In [15], an extensive study of social context-based fake news detection, presenting a hybrid framework, DFN-SCNC, which fills the critical gap of using user comments on Facebook and Instagram platforms, integrating BERT as a contextual feature extractor and Bi-GRU as a bidirectional sequential processing framework. In their study, they created two new publicly available datasets (FNID and FNFD), which were subsequently scraped in 2019-2024 and included

features such as comment id, time stamps, usernames, the number of likes, and the multi-class labels (FNID, 1388 posts and 11726 comments; FNFD, 3299 posts and 185841 comments). The framework was better in its performance with 97% accuracy and 96% F1-score on FNID and 91.6% accuracy and 90% F1-score on FNFD, which outperforms baseline models such as BERT+LSTM, BERT+GRU, Passive Aggressive, Random Forest, and Gradient Boosting classifiers. The study highlighted that incorporation of social context in the form of user comments gives the complementary discriminative information to news content only and deep contextual comprehension of BERT allows processing the short user-generated statements successfully and Bi-GRU allows sequential dependencies due to both forward and backward contexts. Nonetheless, although DFN-SCNC enhances fake news detection with the help of social context integration and the contribution of new datasets, the framework only considers text features but the element of multimodal information (images or videos) is not taken into account and other features of the user profile (post timestamps or sharing patterns) are not employed to add value to the detection. This suggests that future solutions that combine multimodal functionality, examine social dissemination patterns within networks, scale to multilingual scenarios, external knowledge bases, and practical applications on social media will be needed to offer comprehensive multimodal misinformation detection in a variety of content types and user behavior.

The work in [16], examined the performance, consistency, and visualization capacity of LIME and SHAP, two well-known XAI methods, on 6 ML algorithms with Telco Customer Churn and Breast Cancer Wisconsin datasets. The authors also filled important knowledge gaps in comparing systematic XAI methods especially when stability and computational efficiency are considered under varying data characteristics. Their experimental design compared the two approaches by rerunning them multiple times and found that SHAP was more stable with 18 out of 16 consistent influential features on the Telco data, due to its Shapley value calculations that fairly attribute features. Nonetheless, LIME was much faster than SHAP, with explanations taking less than 0.5 seconds against the 1-5 seconds of SHAP. The paper has emphasized a key trade-off between the perturbation-based visualization offered by LIME and the theoretically sound global and local explanations provided by SHAP by several types of visualization, at the cost of computation. The authors have concluded that the choice of methodology depends on the needs of the application- SHAP in trust-critical applications that demand consistency and LIME in application that require high-dimensional data and place a premium on time. Nevertheless, the research has been restricted to two datasets and admitted that both of these approaches are unstable because of the random sampling in calculations.

The work presented in [17] addressed the shortcomings of the traditional Small Language Models (SLMs) such as BERT, introducing the FND-LLM Framework that essentially attempts to integrate the capabilities of SLMs with the logic capabilities of Large Language Models (LLM). Having noticed that the complex misinformation often requires the knowledge of the external world, they developed a multi-branch configuration. SLMs and ResNet are used to extract textual and visual features. Also, a special visual tampering branch is available which detects the manipulation of images. Lastly, a CLIP based cross-modal branch aligns the modalities. The framework has its own LLM branch which does not just name things but also produces the explanation-based features to demonstrate why content could be false. This assists in the interpolation between detection accuracy and interpretability, which is some big deal in the misinformation study. The findings are encouraging as well. The model has positively impacted the result by 0.75%, 6.8%, and 1.3% points in the Weibo, GossipCop, and PolitiFact datasets, respectively. These advantages underscore the importance of the LLM -based reasoning in the process of detecting fake news. However, the tradeoff is real-time performance. Since the system uses an external API of an LLM, it creates latency and poses a privacy risk. In addition to this, the multi-branch architecture is extensive and subsequent iterations indicate the Distilled, lightweight models should be created.

In [18], proposed FMC, a multimodal fake news detector model, a multi-granularity feature fusion and contrastive learning to solve the weaknesses of current frameworks that do not fully utilize multimodal information in different feature granularities. The authors found that the existing approaches were severely lacking in vital aspects, most notably the fact that they used single-granularity extraction and did not take advantage of the emergent pre-trained models. They base their architecture on BERT and ResNet50 to use fine-grained textual and visual features and CLIP to use coarse-grained cross-modal semantic understanding. Multi-granularity fusion module applies MSE loss-based cross-modal alignment and then, by applying textual-visual co-attention, it generates multimodal representations. A multi-head self-attention and supervised contrastive learning joint learning module reduces intra-class distances and increases inter-class separation using a temporal- windowed positive/negative sample selection using cosine similarity. Extensive testing on ReCOvery, GossipCop, and MR2 indicated that FMC was better at accuracy enhancement (1.61-4.65) and MCC enhancement (4.04-4.65) than the existing state-of-the-art algorithms such as TTEC, COOLANT, and MMFN. Multi-granularity features Ablation experiments found that only multi-granularity features become minimally beneficial at unimodal levels and much more beneficial at multimodal levels, and contrastive learning plays a substantial role in learning discriminative representations. However, computational complexity and frozen pre-trained parameters, in real-time deployment, can constrain domain-specific adaptability and scalability.

The recent research [19] created an explainable fake news detector system that overcomes the significant constraints of binary classifications that simplify the evaluation of misinformation by neural networks as black boxes. The authors found that there were inherent gaps in the approach to fake news detection as any other NLP task, failing to consider multi-factor contextual aspects of credibility evaluation. They have a framework created in the SWAROG project in Poland, which uses a 13-question schema of annotation consisting of verification factors (source confirmation, data currency), manipulation factors (cherry-picking, misleading intent), and metaphysical factors (satirical content, political/religious elements). Sentence-BERT features are processed by the pipeline by using two versions, multilingual XLM-RoBERTa and Polish-to-English translation, and One-Hot encoded annotator demographics (age, education, political/religious views, media consumption), which produced an enriched feature vectors. A set of N classifiers give finer responses to questions, which are later combined to score holistic credibility. A study on 40,000 annotated Polish digital media articles reached 72.1% balanced accuracy and a range of news source recognition of 60.9-87.9% in 21 sources. Competitive analysis of PubHealth dataset showed competitive results (87% true news, 78.9% false news recognition) but with a higher explainability factor-based analysis compared to attribution-only approaches. Nonetheless, reliance on annotator subjectivity and the possibility to use explainability mechanisms adversarially are known limitations that should be used with caution.

The recent work in [20] approached the issue of the lack of concrete tools in Dravidian languages by training a large-scale benchmark of various ML, DL, and transformer models to identify fake news and offensive content in Malayalam, actually paid attention to the linguistic complication and code-mixing that is so prevalent on local Indian social media, trying a plethora of architectures between standard SVMs and CNN-LSTMs up to more advanced multilingual transformers such as the mBERT or XLM-RoBERTa. The finding shows that XLM-RoBERTa performed a gangue of fake news with an F1 -score of 99.17, and mBERT was the hero for flagging offensive content (F1 of 95.95) and beat the non-transformer baselines by a significant margin. The study incorporated three XAI techniques, namely LIME, Anchor and Occlusion to visualize decision boundaries, demonstrating for the first time how these techniques can indeed highlight the important tokens in Malayalam writing that makes the predictions in the model. But there are also some gaps as research did not fully dig into the misinformation, which is represented by images in the current time. Also, they considered static, curated datasets, which do not necessarily reflect the reality of the speed and

adversariality of social media streams in reality. This demonstrates that we require the future multimodal frameworks that would introduce the explainability mechanism to image-text pairs in low-resource languages.

The existing research [21] provided a thorough study of misleading content detection approaches in natural language, presenting a two-part system comprising of the CoDE and the DiDA, offers solutions to the most important issues of domain adaptability and the ability to be semantically flexible. The paper highlighted the shortcomings of conventional methods based on the fixed feature extraction or generic pipelines of classifiers that cannot handle domain variations and tend to struggle with vague expressions or weak supervision as well as multi-domain variability, leading to lower generalization and disparate real-life performance. The suggested architecture does not only entail the use of dynamic attention mechanisms on hierarchical semantic encoding and domain-conscious feature modulation to learn adaptive representations in both unimodal and multimodal input, but also incorporates domain priors through contrastive learning and feature retrieval-based regularization to make inferences much more robust even in the case of drastic domain shift. There are Detailed experiments on a heterogeneous dataset i.e., social media posts (FakeNewsNet, WELFake), health misinformation (ReCOvery), and claim verification data (Emergent) shows significant improvements over state-of-the-art baselines were observed with consistent performance across a wide range of linguistic and contextual scenarios. Nonetheless, as the framework promotes detection via its modular framework and weakly supervised training capacities, it remains dependent on the availability of correct domain prior to successful contrastive learning and concentrates on textual content even though it theoretically accepts multimodal inputs (essential to manage rich multimedia). This brings into focus why more adaptive unsupervised domain estimation methods and an enhanced incorporation of visual-textual co-attention mechanisms are necessary to reflect the multi-modality interaction of the various modalities behind modern misinformation strategy in the changing social media ecosystems.

The study [22] addressed the challenge of multimodal noise and weak feature alignment in misinformation detection by proposing GAConF, a novel framework that integrates Gated Attention mechanisms with a unique Category Perception Loss. Recognizing that irrelevant or weakly correlated image-text pairs can confuse traditional fusion models, the authors designed a Gated Attention module to dynamically filter out noisy features before fusion, ensuring that only semantically meaningful cross-modal interactions are preserved. Furthermore, the study introduced a Category Perception Loss function to enforce tighter intra-class compactness and distinct inter-class separation in the feature space, thereby enhancing the model's discriminative power. Experimental evaluations on the Weibo and PHEME datasets demonstrated that GAConF achieves superior performance compared to state-of-the-art baselines by effectively mitigating noise and improving feature representation. However, the study's focus on standard benchmark datasets limits insights into the model's robustness against sophisticated, adversarial manipulations (e.g., deepfakes) or its generalization to languages and platforms beyond those tested, suggesting a need for future research into adversarial robustness and cross-platform transferability.

The recent research [23] addressed the dual challenge of detecting inconsistencies *within* and *across* modalities in fake news by proposing CroMe. Recognizing that previous methods often treated text and image encoding independently or focused solely on their alignment, the authors developed a framework that utilizes BLIP encoders for robust feature extraction, coupled with a Proxy Anchor Metric Learning module to sharpen intra-modal feature discriminability. This is integrated with a Cross-Modal Tri-Transformer that fuses text, image, and combined representations to capture subtle inter-modal conflicts. Experimental results on the Weibo and Weibo-21 datasets demonstrated that CroMe significantly outperforms existing multimodal baselines by effectively identifying both intra-modal and inter-modal content. However, the intricate multi-branch architecture imposes a higher computational burden, and the study's primary validation on specific

Chinese social media datasets suggests a need for further research into its cross-lingual generalization and efficiency in real-time, high-volume news environments.

A thorough study of the methodology of fake news detection was carried out by [24] in which SGAMF was proposed, a framework that seals the gap of critical importance in multimodal fusion; unstructured text features are integrated with structured tabular features. The analysis highlighted the shortage of the extant methods that focus on the fusion of text only and image-text only, but do not consider the possibility of textual information combined with auxiliary data like user profiles, behavioral patterns and tags. The sparse gated attention mechanism suggested is not only able to allow interaction of dynamic features, but also offers selective filtering of irrelevant data, thus enhancing detection of the data, in addition to preserving computational efficiency. The performance on a new set of COVID-19 fake news data showed better performance than the state-of-the-art methods, and their results were competitive in accuracy and the F1 score. Nevertheless, unlike SGAMF that offers multimodal fusion with an architecture based on attention, it engages with the auxiliary features that are manually selected and does not investigate the aspects of visual features integration or the patterns of temporal propagation, which are key to full fake news detection in multimedia-intensive social media settings. This brings up the necessity of more integrated solutions that rely on a range of different sources of information such as images, videos, and network propagation dynamics in order to include the multidimensional nature of the spread of misinformation.

In [25], an extensive study of cross-source multimodal fake news detection techniques, introducing CMFNThinker, a network that is capable of dealing with the essential issue of the deterioration of performance when training on one source data are used to detect news on other platforms. The research highlighted the shortcomings of currently available methods that have a strong dependence on quality labeled information of certain sources and are unable to transfer to various social media platforms with different content structures, styles, and audience features. The suggested three-phase architecture does not only model human cognition processes as perception and observation, reflection and analysis, and reasoning and decision making, but also uses historical news cases and information of social background to offer interpretable evidence in making classification decisions. The performance analysis on multi-source datasets was shown to be higher than those of state-of-the-art through classification of at least 11.3% in macro F1 score in cross-source detection with much lower performance decay rates. Nevertheless, although CMFNThinker moves in the direction of improving cross-source detection due to the use of its case-based reasoning system and natural language inference systems, it has sensitivity to the size of the historical case database, and performance declines with adding multiple similar news posts, and social context is considered as background information, which is important in managing the dynamism of fake news across heterogeneous platforms. This makes it necessary to have more advanced attention mechanisms capable of dynamically balancing various contextual factors and adaptive retrieval strategies capable of maximizing the number and relevance of historical cases depending on the nature of the target news sources.

The recent research in [26] performed an in-depth study of the multimodal fake news detection problem presenting a framework FND-MC, overcomes the shortcomings of current unimodal methods implemented by means of cross-modal alignment and multimodal fusion schemes. The researchers used a multimodal feature-extracting model (CLIP), the text-sensitive model (BERT), and multi-head attention mechanism to match inter-modal associations. Their framework proved to be performing better on two public datasets with 82.24% macro-F1 on ReCOvery and 90.34% on GossipCop which is better than baseline methods like SAFE, COOLANT, TTEC, CAFE, BTIC, SpotFake, and MMFN. The study has highlighted the significance of contrastive learning in aligning text-visual features and also found the role of textual information to be more dominant in the classification decision as compared to the visual features and also BERT is significantly better in unimodal comparisons than ResNet. Nevertheless, though FND-MC enhances multimodal

integration by utilizing complex mechanisms of attention, cross-modal alignment, the framework preserves frozen parameters of pre-trained models during training, which can become a constraint of the framework to adapt to domain-specific patterns. This illustrates the necessity of future solutions that would strike a balance between computational and fine-tuning, especially in the context of tracking the trend of fake news and various multimedia content depending on the social media and cultural environments.

In [27] effective framework is developed to examine the explainable fake news detection problem in detail on social media websites, providing XFND, a hybrid architecture, that can tackle the issue of interpretability of transformer-based black-box models without compromising the classification performance. They used a combination of Distil-BERT contextual embedding extraction and BiLSTM to model in order to process article titles and bodies with tokenization, attention, and fully connected classification layers. The framework was highly accurate at 98% in the Fake and Real News dataset, which was 2% higher than state-of-the-art methods including Logistic Regression (94%), BiLSTM (96%), BERT (95%), Distil-BERT (96%), ALBERT (95%) and T5 (48%), and 2% higher than base models. The methodology used LIME to increase transparency by detecting influential textual features using perturbed input analysis, which produces weighted score of importance, which indicated particular words that positively or negatively affect classification decisions, including fake news detection by Tower and Philippines, and authentic news verification by Tillerson and Russia. Their experimental findings showed a better performance of the hybrid architecture when compared to standalone transformers, and comparative analysis revealed the limitation of T5 as a generative based architecture that is mainly suited to sequence-to-sequence tasks but not classification. Nevertheless, although XFND progresses explainability based on post-hoc interpretation methods and has a strong performance based on bidirectional processing, the research paper recognizes both the limitations related to datasets and the fact that the explanations provided by LIME are inaccurate in some cases. This indicates that a further study is necessary that uses retrieval-augmented generation with large language models such as GPT-4 to improve the performance of detection and the quality of explanations on various areas of misinformation and multilingual settings.

In [28], a comprehensive framework that entails the fine-tuning of BERT and RoBERTa models based on the FakeNewsNet dataset and attempted to deal with the black-box problem with LIME. The experiment used post-hoc explainability to discover influential features to classify, where BERT (98.16) and RoBERTa (99.20) have an accuracy that is 3 points above the state-of-the-art. The framework passes the concatenated news titles and the body text as input to the mechanisms of self-attention and the tasks used in the framework include tokenization, positional embedding, and segment embedding to create contextual representations. LIME improves interpretability by making more relevant text samples by using exponential kernel functions to weight them according to their similarity to the original input and training simpler linear models to make similar predictions locally. The study has shown that LIME is successful in detecting important keywords that impact classification success and that visual explanations with color-coded highlighting (fake news (orange), real news (blue)) can help build user confidence in AI-driven detection systems. Nevertheless, although the framework adds model transparency by means of post-hoc interpretation, the authors admit that LIME does not help in giving valid explanations to all cases, especially those ones where local approximations may not best represent the global decision-making patterns of deep transformers. This demonstrates why hybrid explainability methods that use LIME with other interpretability methods like SHAP should be developed to enable more robust and comprehensive explanations in a variety of linguistic scenarios and multimodal datasets.

In a thorough exploration of stance detection in politics, [29] propose a hybrid model BERT-BiLSTM that combines the best of both deep contextual and sequential dependencies that can be used to effectively determine implicit or indirect stance in social media. They utilized fine-tuned BERT embeddings that were specifically conditioned on Twitter political writing mixed with Bi-

LSTM layers to process tweets of the 2020 U.S. presidential election and measured performance on P-Stance and SRQ (Stance in Replies and Quotes) benchmark datasets using three-class classification (Against, Favor, None). The framework also outperformed traditional ML methods such as SVM (67.69% on P-Stance), RF (69.22%), and standalone BERT (77.80% on P-Stance, 55.17% on SRQ) by achieving 79.60% accuracy, 79.63% P, 79.60% R, and 79.57% F1-score on P-Stance, and 56.13% accuracy and 55.3 The study highlighted that the self-attention system of BERT can well encode rich contextual semantic information and word connections in sentences even though the BiLSTM can encode sequential connections and time pattern essential in interpreting subtle political language with sarcasm, implicit allusions, and comparisons. Nevertheless, although BERT-BiLSTM enhances stance detection by integrating complementary architectures, the framework has high computational costs due to transformer architecture, training data biases against generalization in different political settings, difficulties in detecting stances that are realized via sarcasm or non-literal language, and an imbalanced distribution between the data in the dataset and the class of Nobody. This shows how future strategies should consider domain adaptation methods to cross context transferability, scale up datasets with multilingual political topics and international news, add other social media capabilities like likes, retweets, and follower networks to capture more engagement patterns, and improve model interpretability to gain insight into how decisions are made in politically sensitive app uses.

A detailed rumor-detecting study by [30] of social networks proposed a model, which supports rumor propagation-level features (GCNs) and source-level features (GNNs) hence overcoming drawbacks in understanding semantic innuendoes and full retweet propagation patterns. In their work, they tested on the Sina Weibo dataset of 4,664 events (2,351 rumors, 2,313 non-rumors) and their results showed that they obtain a high accuracy of 97% with 2% better than the state-of-the-art baselines such as HAGNN (95%), VAE-GCN (94%) and RvNN (90%). The framework showed that GCN is useful in aggregating features of the neighbors of the event graphs that are built on source tweets and 20 retweets, whereas GNN with GRU works with word-level relationships, utilizing reset and update gates that select the irrelevant information and balance the contextual information. Nevertheless, as DGBM scales rumor detecting by dual graph architecture and Gated-GNN integration of local and global features, limitations of computational resources only made 20 retweets per event, dependence on Sina Weibo as the platform because twitter dataset did not provide complete retweet texts to process with BERT, and lack of multimodal features or user profile features. This demonstrates the necessity of novel methods that combine real-time detection features, the ability to operate across platforms, the systematic optimization of hyperparameters, and the addition of new features such as emotion analysis and multimodal content processing to filter the misinformation in their totality.

The study in [31] presented FACT-CLIP a multimodal fake news detector that uses CLIP-based cross-modal attention and transformer fusion to overcome impediments of the current systems. The authors found that the previous techniques poorly exploited the visual data, where the main reliance was on CNN-based networks, which only represented holistic images and lacked entity-specific data which was vital in identifying discrepancies between text and images. The system removes two level features of visual information: global image features using ResNet-152, and entity-level features using Faster R-CNN, and textual features using BERT. FACT-CLIP uses the pre-trained multimodal encoder of CLIP to produce cross-modal alignment through their semantically rich representations in the form of text-image pairs and calculates the cosine similarity scores. The framework uses an attention mechanism which selectively weights features of text, images and CLIP embeddings, and then a custom transformer encoder with multi-head self-attention to get deep modal-modal semantic connections. Experimental testing on the Twitter, MediaEval dataset indicated that FACT-CLIP was more accurate by 2.8% than baseline methods and better than other methods, including EANN, SpotFake, and BDANN. Nevertheless, the paper recognizes the computational constraints since the large-scale transformer model requires a lot of GPU power in

contrast to conventional CNN methods. Also, the framework was tested on a single dataset because of resource limitations and the authors state that the fact that CLIP depends on image-text correlations can lead to strong similarity scores where contextually related and fabricated content appear. Further studies are needed in the future to use bigger datasets and strengthen the explainability mechanisms to improve reliability in real-world operational cases.

The key innovation in [32] created an explainable fake news detector based on the DL ensemble and source credibility verification to overcome the transparency shortcoming of the conventional black-box models. The authors found that current detection systems had severe shortcomings, especially in how they are opaque in decision-making and they do not adequately consider source verification. Their architecture is based on three neural networks; CNN to find local patterns, LSTM to find sequential dependencies, and BiLSTM to find bidirectional contexts whose predictions are aggregated by averaging of probabilities. It is a new source verification module that uses LLaMA3-70B over Groq API to test the reliability of content origin independently using generative AI. The system was tested on a balanced dataset (190,000 news articles) with an accuracy of 95.2 and BiLSTM showed better results individually with 92.7% accuracy than LSTM (91.2%) and CNN (89.4%). The explainability framework offers model confidence visuals, highlighting of feature importance and feature verification explanations using an intuitive web interface. Improvement with ensemble, however, was very slight (0.4%) and with a huge computational cost, implying that BiLSTM might provide the best performance-efficiency balance in production. The key constraints are that a resource intensive architecture and high-latency API dependencies imply scalability limitations and this issue should be addressed by proposing model compression methods and alternative source verification solutions in real-time processing applications.

In [33], a framework SR-CIBN was proposed that goes beyond simple feature fusion to address the nuanced challenge of detecting multimodal fake news where visual and textual elements are deceptively consistent, aiming to mislead detection systems that rely solely on obvious discrepancies. The model employs hierarchical penetration to interact with multimodal features aligned via contrastive learning, extracting both global consistency and inconsistency signals. A key innovation is the dynamic balancing mechanism that adjusts the fusion intensity of these signals based on the degree of image-text matching, ensuring that the model focuses on the most discriminative features for a given news item. Furthermore, the integration of a triplet loss based on semantic relationships effectively prevents confusion between semantically similar real and fake news. Extensive experiments on Twitter and Weibo datasets demonstrated that SR-CIBN outperforms comparable state-of-the-art approaches, successfully identifying sophisticated forgeries even when image-text pairs appear superficially aligned. However, the reliance on intricate balancing mechanisms and triplet loss training suggests a high computational overhead, and the study's focus on standard benchmarks leaves open questions about its robustness against emerging, highly adversarial deepfake content that might bypass current consistency checks.

The framework in [34] have made a full-scale deep exploration of explainable multimodal fake-news detection and released a completely new framework, which consists of AutoML analysis on images with prompt-enhanced Large Language Models on text, all connected through Functional Networks to handle the typical headaches of cross-modal connectivity and interpretability. The study note that most current approaches are more like black-box systems, using massive conveniently-labeled data and extensive fine-tuning, in which case a complete pain in low-resource settings where fake news is typically under-labeled to be fiddled with on a domain-specific basis. The two pipelines design was very clean as it has Fuzzy C Means clustering with AutoGluon to do solid image classification (with 95.26% acc. on the FIDAC dataset) and a ChatGPT based prompt trick to do text classification, getting 92.41% acc. on the Twitter dataset. Functional Networks is a mathematically transparent, stable fusion engine which dynamically weights every modality and makes things explainable via LIME. That notwithstanding, although the framework takes the detection to the next level by pushing forward with a comprehensive XAI blend of feature

importance diagnostics using SHAP and axiomatic checks of identity, separability, and stability, it nevertheless contains some cracks. It fits to predetermined fuse weights on diverse data sets, which can cause it to forfeit a modality-specific advantage when content is sparse, it uses proprietary LLMs like ChatGPT, such that its images of reproducibility are limited even when claimed by the authors to be open-source, and it may slightly lag on image alone models such as BuzzFeed or PolitiFact. These are some of the main pitfalls in an attempt to nail down the optimal adaptive fusion in a quagmire of social media where the quality of texts and visual manipulation, as well as cross-modal consistency, wildly vary between platforms, types of content and tactical evolution of misinformation.

Recent study [35] have proposed a solution of addressing the issues of English-centric and opaque misinformation-detection systems by their new framework HEMT-Fake, a hybrid explainable multi-mode transformer architecture. The article presented a huge dataset of 74,000 English news articles and four low-resource Indian languages: Hindi, Gujarati, Marathi, and Telugu, and combines CNN-BiLSTM with text, ResNet50 with images, and GraphSAGE with metadata. The proposed architecture is a combination of hierarchical explainability with SHAP and LIME where you can see what is going on at token, sentence and modality levels. Besides, it is more competitive by approximately 5% increase in Macro-F1 compared with the basic models such as XLM-R. Nevertheless, HEMT-Fake is a significant improvement over the low-resource detection, with an inference latency of roughly 140ms per article, the inference time is still reasonable to use in fact-checking in real time, although it is limited to languages spoken in India. This actually underscores the importance of optimization tips to strike the right balance between profound interpretability and the pace requirements of social media surveillance (monitoring in real-time).

3.1 Gap Analysis

Table 3.1: Gap Analysis of Existing Review

Ref.	Proposed Model	Models Used	Dataset	Acc.	Evaluation Metrics	Strength	Limitation	Gaps
[1]	dEFEND	HAN, Co-Attention	PolitiFact, GossipCop	+5% F1	Precision, Recall, F1	Social explainability, crowd-sourced verification	Cold-start problem, text-only	Visual + social fusion
[2]	Multi-image FND	BERT, VGG-16, LSTM, Cosine Similarity	GossipCop	+5.39% F1 over SpotFake	Acc, Precision, Recall, F1	Sequential multi-image visual features, text-image similarity	Domain-specific, no deepfake detection, under-sampling	Image forensics, cross-domain generalization, deepfake robustness
[3]	FND-CLIP	ResNet, BERT, CLIP	Weibo, PolitiFact, GossipCop	Up to +6.8%	Acc, Precision, Recall, F1	Excellent cross-modal alignment, dynamic weighting	Dataset/domain limited, pre-trained model dependency	Domain-adaptive CLIP tuning
[4]	SSM	Hype-HAN, TextRank, EfficientNetB7, BiGRU, ELA	FNDD, TI-CNN, FNSDS	96.74%, 96.50%, 96.90%	Acc, Precision, Recall, F1, ROC-AUC	Five-module framework, stylometric + semantic analysis, image forgery detection	Computational complexity, manual threshold tuning, English-only	Real-time deployment, cross-lingual generalization
[5]		Cross-Attention,	AI + Human Mix (custom)	+10.2 F1 points	F1	AI-generated fake focus, visual	Limited dataset	Standard benchmark testing, broader

		GradCAM, Attention Maps				+ textual explainability	validation, technical report only	generalization
[6]	DANN+LIME	DANN, Conv1D, LSTM, GloVe, LIME	CoAID (4.5k), MiSoVac (190k COVID-19)	+3% Acc, +9% AUC	Acc, AUC, F1	Cross-platform domain adaptation, unsupervised, explainable	LIME instability, limited to 2 datasets	Multimodal analysis, federated learning
[7]	ExFake	Content + Social Context + Knowledge Graph	Real-world datasets	Improved SOTA	Macro-Acc, Macro-Precision, Macro-Recall, Macro-F1	User-centric explanations, fact-checking integration	Data dependency, cold-start problem, real-time scalability	Autonomous zero-shot XAI systems
[8]	TELLER	LLM + Neuro-Symbolic (dual-process)	Multi-domain (4 benchmarks)	SOTA-level	Acc, F1	Controllable, explainable reasoning paths, generalizable	Manual predicate design, LLM factual dependency	Auto predicate generation, neuro-symbolic self-correction
[23]	CNN-LSTM + FastText	CNN-LSTM Hybrid, FastText, BERT, XLNet, RoBERTa, LIME, LDA	WELFake, FakeNewsNet, FakeNewsPrediction	0.99, 0.97, 0.99	Acc, Precision, Recall, F1	OOV handling, L2 regularization, LIME+LDA explainability	Text-only, no visual integration, English only	True multimodal fusion, multilingual models, global XAI (SHAP)
[10]	Tri-FusionDet	DistilBERT, ResNet18, XGBoost, Cross-Attention	Fakeddit (682k)	94% (+6% over baseline)	Acc, F1, Precision, Recall	Late fusion of text, visual, social engagement features	Pre-trained model dependency, no domain-specific engineering	Advanced attention, explainability techniques
[11]	BC-FND	CLIP (Text/Image), BERT, Hierarchical Bilinear Fusion	Weibo16, WeChat18, Weibo23	94.2% (Weibo16), 90.0% (Weibo23)	Acc, Precision, Recall, F1	CLIP alignment in shared embedding space, consistency loss, handles document images	Temporal shift degradation (5.5% drop)	Time-invariant learning, external event verification
[12]	MEFaND	BERT, GraphSAGE, GAT, VGG19	PolitiFact, GossipCop	99% (PolitiFact), 96% (GossipCop)	Acc, Precision, Recall, F1, AUC	Early detection (5h), node+edge enhanced graphs	No visual modality, domain-specific limitations	Visual integration, multilingual support, ethical bias mitigation
[13]	RAFAH	RoBERTa, Hybrid features (Linguistic, Psycholinguistic, Emotional, Readability)	BuzzFeed, ISOT	89% (BuzzFeed), 99% (ISOT)	Acc, Precision, Recall, F1	Multi-dimensional feature engineering, superior to baselines	Text-only, no multimodal, no social propagation	Multimodal integration, social propagation, multilingual
[14]	Pyramidal Co-Attention	PRU, HAN, Sentence-Comment Co-Attention	FakeNewsNet (PolitiFact, GossipCop)	+5.56% F1, +16.4% NDCG	Acc, Pre, Rec, MAP, NDCG	Explainability, check-worthy sentences, human-validated top-k comments	Relies on ClaimBuster, no external knowledge base	Direct fact-checking integration, trustworthiness assessment
[15]	DFN-SCNC	BERT, Bi-GRU, Social Context (Comments)	FNID (Instagram), FNFD (Facebook)	97% (FNID), 91.6% (FNFD)	Acc, Precision, Recall, F1	Novel datasets, social context integration, user comments	Text-only, no user profile features, no temporal patterns	Multimodal features, social propagation, real-world deployment

[16]	LIME vs SHAP	LIME, SHAP, NB, LR, RF, Gradient Boosting, DT, MLP	Telco Churn (7k), Breast Cancer (569)	N/A	Accuracy, Consistency, Runtime	LIME: faster (0.5s); SHAP: more consistent and stable	LIME: unstable (random sampling); SHAP: high computation	Not fake news detection; XAI comparison only
[17]	FND-LLM	SLMs, ResNet, CLIP, LLM (multi-branch)	Weibo, GossipCop, PolitiFact	Up to +6.8%	Acc, Precision, Recall, F1	Reasoning-based explanations, visual tampering detection	LLM latency, privacy risk, complex architecture	Distilled on-device LLMs
[18]	FMC	BERT, ResNet50, CLIP, Multi-head Attention, Contrastive Learning	ReCOvery, GossipCop, MR2-E, MR2-C	85.48%–92.03%	Acc, Precision, Recall, F1, MCC	Multi-granularity fusion, cross-modal alignment, contrastive learning	Computational complexity, frozen pre-trained parameters	Real-time deployment, domain-specific adaptability
[19]	SWAROG	Sentence-BERT (XLM-RoBERTa), One-Hot Encoding, Ensemble classifiers	Polish digital media (40k), PubHealth	72.1% BAC	BAC, Precision, Recall, F1	13-factor explainability, annotator demographics integration	Annotator subjectivity, adversarial vulnerability	Binary aggregation from multi-factor, generative AI integration
[20]	Malayalam XAI	mBERT, XLM-RoBERTa, LIME, Anchor, Occlusion	Malayalam dataset	F1 = 99.17% (fake news)	Acc, Precision, Recall, F1, 10-fold CV	Low-resource language focus, XAI token visualization	Text-only, static curated datasets	Multimodal low-resource XAI
[21]	CoDE-DiDA	Transformer, Multi-head Attention, Memory Bank, Contrastive Learning	FakeNewsNet, WELFake, ReCOvery, Emergent	94.82% (FakeNewsNet), 93.71% (WELFake)	Acc, Recall, F1, AUC	Domain adaptation, weak supervision, hierarchical encoding	Depends on domain priors, primarily textual focus	Deeper visual-textual co-attention, unsupervised domain estimation
[22]	GAConF	Gated Attention + Category Perception Loss	Weibo, PHEME	Improved SOTA	Acc, F1	Noise reduction, intra/inter-class separation	No adversarial testing, limited platform scope	Deepfake robustness, cross-platform transferability
[28]	CroMe	BLIP, Proxy Anchor Metric Learning, Cross-Modal Tri-Transformer	Weibo, Weibo-21	Strong gains over baselines	Acc, F1	Captures intra + inter-modal conflicts	Heavy computation, Chinese-platform specific	Cross-lingual generalization, real-time efficiency
[24]	SGAMEF	Improved ALBERT, Sparse Gated Attention, Entity Embedding, BiLSTM	Weibo COVID-19, CHECKED	94.2% (Weibo16)	Acc, Precision, Recall, F1	Sparse attention filters noise, balances accuracy and efficiency	No visual modality, manually selected features, temporal degradation	Time-invariant features, external knowledge verification
[25]	CMFNTinker	BERT, KeyBERT, VGG19, Sentence-BERT, Cross-	Weibo-21, MCFEND (13 sources)	90.2% (Weibo-21), 58.0% (MCFEN	Macro-F1, Decrease Rate	Cross-source detection (+11.3% F1), human-thinking simulation	Database size sensitivity, performance drops with similar posts	Dynamic attention weighting, adaptive retrieval strategies

		Attention		D-G1)				
[26]	FND-MC	CLIP, BERT, Multi-head Attention, Cross-Modal Alignment	ReCOvery, GossipCop	82.24% F1 (ReCOvery), 90.34% F1 (GossipCop)	Acc, Precision, F1, Recall	Multimodal fusion, cross-modal alignment, fine-grained features	Frozen pre-trained parameters, no fine-tuning adaptation	Domain-specific fine-tuning, real-time detection
[29]	XFND	DistilBERT, BiLSTM, LIME	Fake and Real News dataset	98%	Acc, Precision, Recall, F1	Hybrid architecture, interpretable LIME features, 2% over SOTA	LIME inaccurate in some cases, dataset limitations	RAG + GPT-4 for explanations, multilingual settings
[30]	BERT + RoBERTa + LIME	BERT, RoBERTa, LIME	FakeNewsNet	98.16% (BERT), 99.20% (RoBERTa)	Acc, Precision, Recall, F1	Post-hoc explainability, high accuracy, color-coded visualizations	LIME local approximation limitations, black-box nature	Hybrid explainability (LIME+SHAP), multimodal datasets
[31]	BERT-BiLSTM	BERT, BiLSTM, Fine-Tuning	P-Stance, SRQ	79.60% (P-Stance), 56.13% (SRQ)	Acc, Precision, Recall, F1	Contextual + sequential features, political stance detection	High computational cost, sarcasm detection challenges, data bias	Domain adaptation, multilingual support, social features
[32]	DGBM (Dual GNN-BERT)	GCN, GNN (GRU), Gated-GNN, BERT	Sina Weibo (4,664 events)	97% (+2% over HAGNN)	Acc, Precision, Recall, F1	Dual graph architecture, local + global propagation features	Limited to 20 retweets, platform-specific, no multimodal	Real-time detection, cross-platform, emotional analysis
[33]	FACT-CLIP	BERT, ResNet-152, Faster R-CNN, CLIP	Twitter MediaEval (13k)	85.8%	Acc, Precision, Recall, F1	Two-level visual features, cross-modal attention, entity-level features	Computationally expensive, single dataset evaluation	Real-time scalability, stronger explainability mechanisms
[34]	Explainable FND	CNN, LSTM, BiLSTM, LLaMA3-70B (Groq API)	HuggingFace, FakeNewsNet (190k)	95.2%	Acc, Precision, Recall, F1	Source verification with generative AI, multi-visualization explainability	Ensemble overhead (0.4% gain), high-latency API, scalability issues	Model compression, alternative source verification
[35]	SR-CIBN	Contrastive Learning, Triplet Loss, Hierarchical Fusion	Twitter, Weibo	SOTA	Acc, F1	Handles subtle image-text consistency, dynamic balancing mechanism	High computational overhead, standard benchmarks only	Adversarial robustness, deepfake content
[19]	AutoML-Prompt Fusion	AutoGluon, ChatGPT-4.0, FCM, Functional Networks, LIME, SHAP	Twitter, PolitiFact, BuzzFeed (multimodal)	96.7% (T), 95.1% (P), 94.7% (B)	Acc, Precision, Recall, Training Time	Comprehensive XAI (LIME+SHAP), prompt-based low-resource learning	Fixed fusion weights, ChatGPT dependency, reproducibility issues	Dynamic adaptive fusion, open-source LLMs, dataset-aware weighting
[35]	HEMT-Fake	CNN-BiLSTM, ResNet50, GraphSAGE, SHAP, LIME	Indic Multilingual (74k)	+5% Macro-F1 over XLM-R	Acc, F1	Strong explainability, low-resource multilingual focus, 140ms latency	High latency, region-specific (Indian languages only)	Lightweight explainable models, broader language coverage

4 PROPOSED METHODOLOGY

The current explainable multimodal fake-news-detecting system is developed based on a number of critical findings obtained in the field of recent scholarship. It has been demonstrated in previous studies that the combination of textual and visual presentations of features is more effective as compared to the performance of analysis of either presentation. An example is that the FND-CLIP approach has shown an improvement in accuracy between 0.7 - 6.8% with the use of CLIP -based models as compared to traditional convolutional neural network (CNN) models. In a similar manner, research studies characterizing the process of attention-driven fusion have suggested that cross-modal interactive processes cannot be excluded as far as understanding the manner in which textual and visual information can interact to propagate misinformation.

The multimodal fake-news detection system presented in this chapter is based on three converging findings from the literature: (i) The backbone models selected for training are a critical lever for the performance of the system: RoBERTa and CLIP are both trained on web-scale data, and transfer well to the noisy social media text and image that define platforms like Reddit; (ii) The fusion strategy needs to capture inter-modal relationships, not just concatenate feature vectors: the semantic tension between text and image is the primary manipulation surface in multimodal disinformation; and (iii) Explainability must be built into the system, not added as an afterthought: post-hoc attribution applied to a non-interpretable fusion mechanism will produce unreliable and sometimes misleading explanations.

The design attempts to balance three key objectives: competitive classification accuracy, computational efficiency given the constraints of using only a single Google Colab T4 GPU (16 GB VRAM), and substantive, multi-level explanations for technical and non-technical stakeholders. These heavier methods: Zero-Shot LLM, Video-level reasoning, Graph neural networks (GNN) over social propagation graphs were carefully avoided in favor of efficient transformer-based encoders trained under limited resources. This scoping decision is well justifiable from a methodological perspective because it separates the contribution from the core value of the text-image fusion mechanism from other architectural factors.

4.1 Research Design and Procedures

The methodological framework came out through a critical review of the merits and demerits of the approaches, which had been in existence. The study highlights that many earlier systems were very predictively faithful, but stayed very sparse in their explanatory understanding of their decision processes, whereas others studies were very explainably oriented, and thus very sparse in their performance. Our strategy tried to prevent both extremes in our strategy. The research is based on a supervised-learning framework broken down into four stages, each of which takes the research from raw data to evaluated, interpretable predictions:

Stage 1: Data preparation: acquisition of Fakeddit corpus, parallel download of images, filtering for broken-URLs, CLIP normalization, RoBERTa tokenization, packing of the data into HDF5 files, and train/validation/test partitioning in a ratio of 70/15/15.

Stage 2: Model development: a two-stage fine-tuning process, involving a head warm-up and encoder unfreezing, that both adapts the model to the task and reduces the computational burden and catastrophic forgetting.

Stage 3: Integration of stage 2 explainability results: building a framework of 3 levels of explainability (intrinsic cross-attention visualization, SHAP partition-based attribution, and LIME local linear surrogate) with complementary explanations that are computed at different costs.

The staged fine-tuning regime in Stage 2 is important: it resists overfitting by allowing the randomly initialized classification head to converge before encoder weights and perturbed and it enables diagnostic inspection of model behavior at intermediate checkpoints, both desirable properties under a limited GPU-time budget.

4.2 Data Sources And Sampling Procedures

The Fakeddit corpus, which consists of more than one million text-image pairs that were collected on Reddit, is our main data source. The choice of this dataset is based on a number of cogent reasons. First, it has a large size to train deep learning models without overfitting. Secondly, it carries original multimodal content where the textual and visual information collaborates to give truth-value decisions. Thirdly, the data is based on real social-media posts and not on the synthetic ones, which guarantees that representative misinformation trends are represented.

The Fakeddit corpus provides a set of features of each example as an RGB image, an original headline text, and a cleaned variant of a headline, as well as a variety of choices of labels in a variety of classification tasks. To keep the scope of the problem clear, we considered the binary classification problem (real versus fake) to apply to the scope of the current study. None of them also omits ancillary metadata, like the source of the subreddit, the vote score, the commentary of users, and so on, which we flagged as optional predictors in future improvements. For all three splits, sampling is stratified across both classes; data shuffling is done for each training epoch; a fixed random seed (42) is used throughout all of Python, NumPy, PyTorch and CUDA for full reproducibility.

The original plan was for about 100,000 samples. All reported experiments are performed on the final processed corpus of 17,149 samples, which is a result of parallel image downloading, broken URLs filtering and HDF5 packing. The class distribution is 8,716 real (50.83%) and 8,433 fake (49.17%), split into 12,003 training / 2,573 validation / 2,573 test samples. All the results that are reported in Chapter 6 are calculated on this corpus.

Table 4.1: Fakeddit Dataset Detail

Attribute	Value/Detail	Attribute	Value/Detail
Full Name	Fakeddit, A New Multimodal Benchmark Dataset for Fine-Grained Fake News Detection	Corpus Size: Total posts (full)	1,063,106 unique Reddit posts (>1 million)
Authors	Kai Nakamura, Sharon Levy, William Yang Wang	HEMT-CLIP subset	17,149 samples 8,716 real (50.83%) 8,433 fake (49.17%) near-balanced
Year published	2020	Train / Val / Test split	Train 12,003 (70%), Val 2,573 (15%), Test 2,572 (15%) Stratified split · seed 42 · shuffled each epoch
Source platform	Reddit (community-sourced posts across 28+ subreddits)	Number of subreddits	28+ subreddits (spanning news, satire, propaganda, and factual communities)
Data modalities	Text (headline / title) Image (post thumbnail) Metadata (subreddit, score, comments)	2-way label	0 – Real, 1 – Fake Used in HEMT-CLIP. Binary classification task.

Task type	Multi-class and binary fake-news classification	3-way label	True Satire / Parody Fake
Primary Language	English	6-way label	True Satire Misleading, Content Imposter Content Manipulated, content Fabricated content
License/access	Publicly available via GitHub github.com/entitize/Fakeddit	Label derivation	Subreddit-level annotation (not per-post human judgement)

Two corpus statistics are of special importance for further design considerations and are given here as an empirical reason:

- **Title Length (RoBERTa tokenizer):** The median of the titles is 10 tokens in length, the 99th percentile is 35 tokens, and the longest is 71 tokens. With a cap of 128 tokens, 0% of titles are capped. This brevity is theoretically important because in the absence of the larger discriminative cue from the title, the image channel (which carries a richer spatial information) is very important in this corpus (empirically confirmed in Chapter 6, where the image-only AUC is higher than the text-only AUC by 2.79 percentage points).
- **CLIP Similarity Distribution (α):** The cosine similarity of the text and image embeddings of the corpus has a mean of 0.276, a standard deviation of 0.054, and a range of [0.086, 0.490] with no outliers. *Importantly, the mean α for fake posts is higher than for real posts (+0.300 vs +0.253, $\Delta = +0.048 \approx 1$ standard deviation).* This is counter-intuitive, as one would expect manipulated posts to have lower semantic alignment between text and image; however, as stated, it is attributed to the nature of Fakeddit: satirical and mislabeled posts are likely to pair a dramatic and emotional caption with on-topic imagery (high alignment) whereas genuine news posts are more likely to pair a specific headline with a generic stock photograph (lower alignment). The distributions intersect significantly, making α alone a rather weak predictor, but the consistent direction of its relationship with the label makes the CLIP-similarity-gated fusion design mechanistically interesting. The gate learns a soft rule (not hard, 'low α implies fake') and the downstream classifier 'learns' the remaining structure.

4.3 Workflow of the system

The HEMT-CLIP system operates a dual-encoder architecture through a cross-attention fusion module. both textual and visual content are processed by a dual- encoder structure which is then combined through cross attention. The entire workflow of the system is shown in Figure 4.1, showing the data flow from raw input through both encoder paths, the cross-attention fusion with α -gating, and the classification and explainability outputs.

4.3.1 Architecture Components

Text Encoder

The text encoder is RoBERTa-base, a more powerful variant of BERT trained to become robustly optimized to the task by removing the next-sentence prediction objective, increasing batch training and dynamically masking the text. RoBERTa-base contains 125-million parameters. The RoBERTa vocabulary defined by the byte-pair encoding is used for post titles, which are tokenized up to 128 tokens, padded and truncated if necessary and automatically filled with [CLS] and [SEP] boundary tokens. All the embedding layer and the first 8 out of 12 transformer layers are frozen (requires_grad = False) during the training, to avoid catastrophic forgetting of the general-purpose

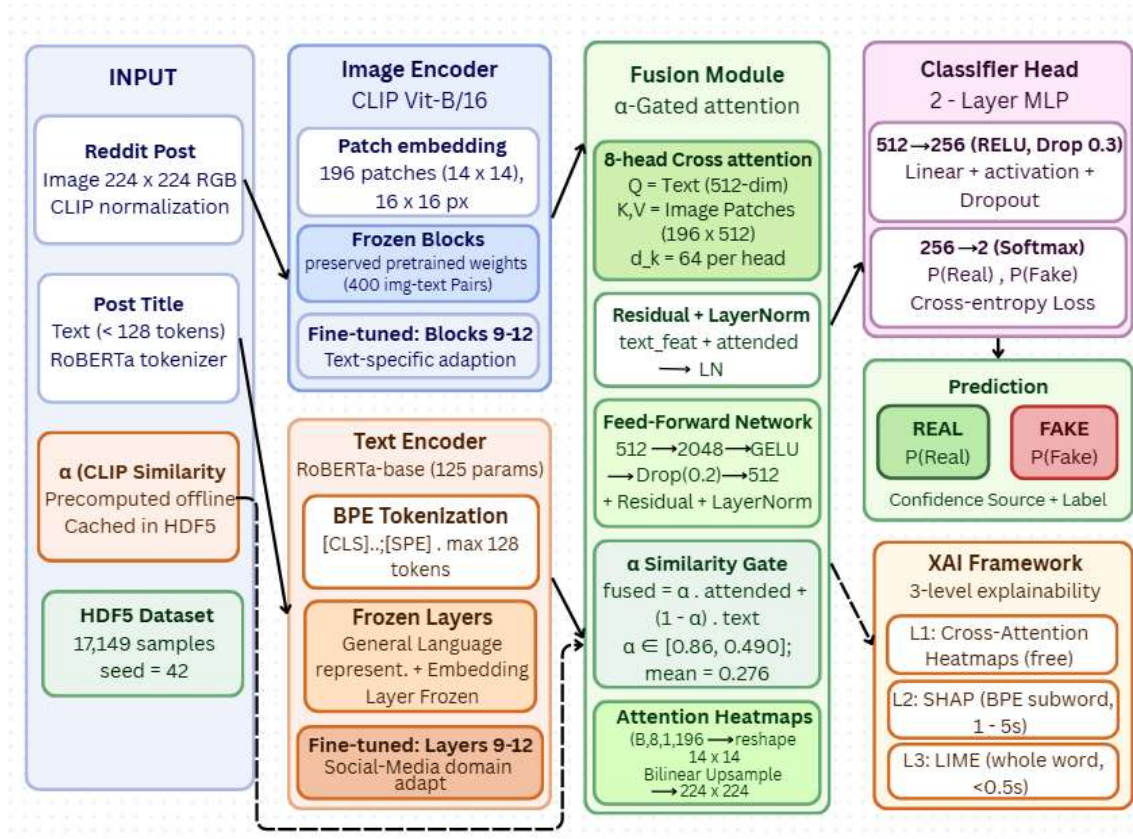


Figure 4.1: System Architecture Diagram

language representations learned during the RoBERTa pre-training stage, and significantly reduce the number of gradients computed. In the final four layers (layers 9-12), the encoder is fine-tuned in Stage 2 to reflect the stylistic and lexical distributions of social-media headlines, enabling the encoder to adjust its contextual representations to align with the style and vocabulary of social-media headlines. At the end of the 12th layer the [CLS] token representation is fed into a projection head (Linear (768, 512) → Layer-Norm → Dropout (0.1)) to get a 512-dimensional text embedding. The projection head can be trained from Stage 1.

Image Encoder

The image encoder is a vision tower designed by CLIP ViT-B/16, which is pre-trained on 400 million image-text pairs collected from the web. Input images are resized to 224×224 pixels before using CLIP's own normalization parameters to normalize them. A learnable [CLS] token is added in front of each image and learned position embeddings are appended to the 197-token sequence before being fed into the ViT transformer stack, with each image subdivided into a 14×14 grid of non-overlapping patches of 16×16 pixels, and a total of 196 patch tokens. The patch embedding layer, the position embeddings and the first 8 (out of 12) transformer blocks are frozen during training. The final four transformer blocks are tuned along with the RoBERTa layers, which are a symmetric design, thus providing equal levels of task adaptation for both the encoders. The encoded output consists of two elements: the output of the pooled CLS vector (512-dimensional vector, semantics of the image in the global context) and the output of the entire 196-dimensional sequence of patch-token vectors (each projected to 512 dimensions, which are used as the Key and Value sequences in the cross-attention computation).

Fusion Module

The fusion module fuses the 512-dimensional text embedding and the 196×512 image patch sequence with an 8-head cross-attention mechanism in which the text embedding is used as the Query and the image patch tokens are used as Keys and Values. This asymmetry is designed: By placing the text as the query, the text-fusion block asks, for every dimension of the textual representation, which image regions are semantically relevant, thus performing query-driven localization without any additional computational complexity, while still providing explanations of the architecture itself.

The eight attention heads attend to a $d_k = 64$ -dimensional subspace of the 512-dimensional embedding, with heads allowed to specialize on different patterns of cross-modal alignment (entity correspondence, tonal concordance, semantic cohesion, spatial co-reference). The per-head attention weight tensors of shape $(B, 8, 1, 196)$ are retained after each forward pass, but are bilinearly unsampled to shape $(B, 14, 14)$, and attention heatmaps of shape 14×14 are stored, at no extra cost to the forward pass, for every head in the model, as described in §4.2.3 and §6.4.

This is then followed by a residual connection and layer normalization, and a second residual connection and layer normalization along with a feed-forward network (FFN: $512 \rightarrow 2048 \rightarrow \text{GELU} \rightarrow \text{Dropout}(0.2) \rightarrow 512$). The attended representation is then gated by the cosine similarity scalar α , as in the gating equation below:

$$fused = \alpha . attended + (1 - \alpha) . text$$

The architectural contribution of HEMT-CLIP is this fusion, similar to the CLIP, but coupled with a similarity gate. If α is high (strong text–image semantic alignment), the gate will give more weight to the cross-attended image-informed representation, and if α is low, the gate will be pulled back towards the cross-attended text representation. The gate does not add an extra set of learnable weights, and thus provides a pure architectural advantage (as measured in Chapter 6) without any capacity cost. The output of this gated fused vector is fed directly into the classification head and has a dimension size of 512.

4.3.2 Processing Pipelines

Figure 4.2, which shows the flow of raw input to the end prediction by the processing stages.

4.3.2.1 Core Mathematical Operations

Cross-Attention Computation: This is achieved with the standard scaled dot-product attention mechanism, where Q = text features (shape $B \times 1 \times 512$), $K = V$ = image patch features (shape $B \times 196 \times 512$) and head dimension $d_k = 64$.

$$Attention(Q, K, V) = \left(\frac{QK^T}{\sqrt{d_k}} \right) \quad Eq. 1$$

Scalar α , Cosine Similarity of embedding of the text of each of the samples with the embedding of the image for each sample. It is precomputed once offline for all 17,149 samples using the full CLIP ViT-B/16 model and cached in HDF5 dataset to avoid repeated computation at every training step: The CLIP similarity α is a scalar per sample computed as the cosine similarity between CLIP's text and image embeddings. It is precomputed once offline for all 17,149 samples using the full CLIP

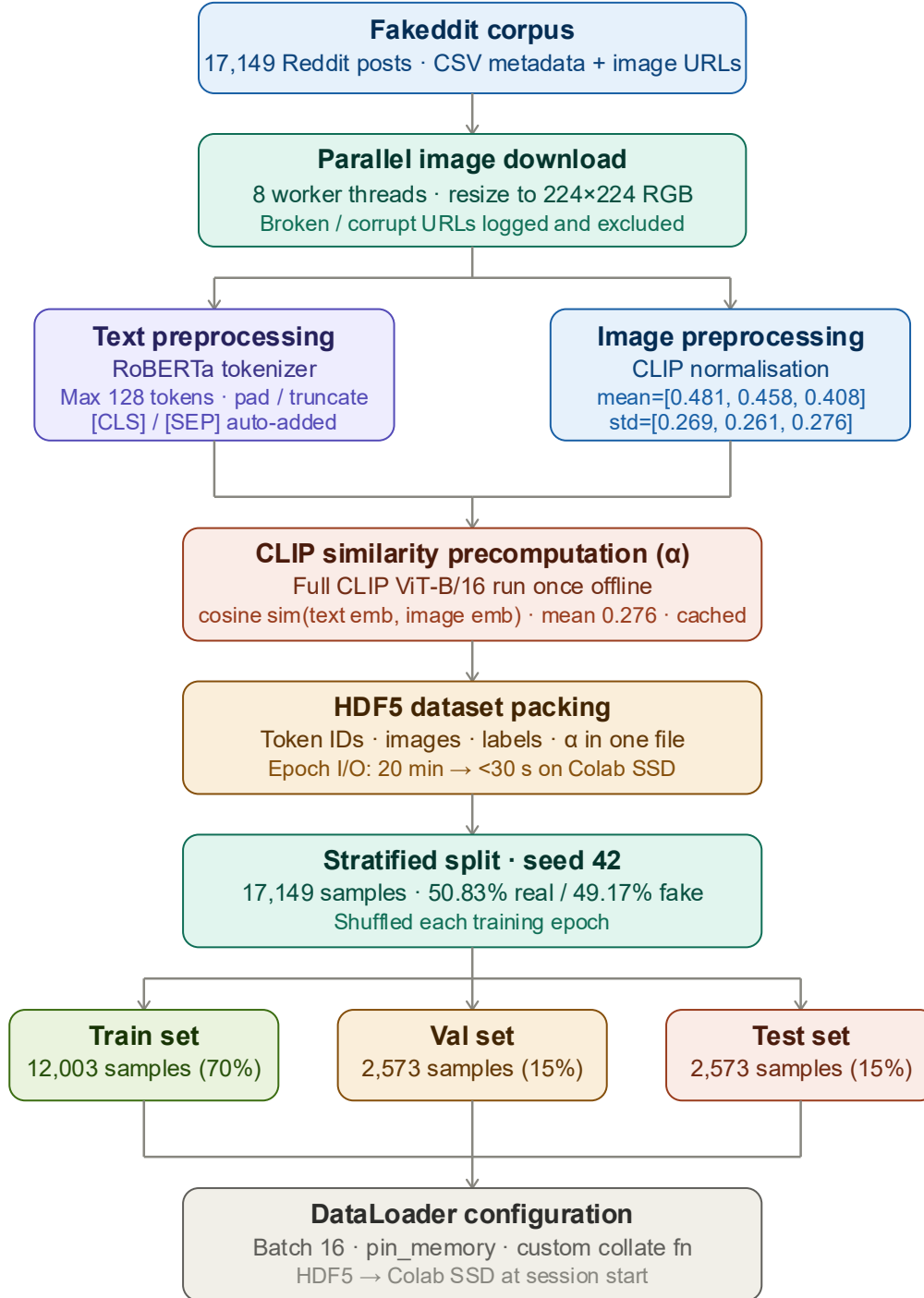


Figure 4.2: Data Processing Pipeline

ViT-B/16 model and cached in the HDF5 dataset to avoid repeated computation at every training step:

$$CLIP_{text}(t) = \cos(\alpha) = \cos(CLIP_{Image}(i)) \quad Eq. 2$$

The CLIP-guided weighted fusion applies the α -gate to produce the final fused representation:

CLIP ViT-B/16:

$$fused = \alpha . attended + (1 - \alpha) . text \quad Eq. 3$$

The full CLIP ViT-B/16 model (vision and language towers) is used to generate a single α value for the entire dataset, reducing the computational cost from a triple to just a single inference. When training and inferring, the value of α is loaded from the HDF5 file and inserted directly into the fusion gate at the same time with no additional I/O overhead. This caching technique allows for a T4 budget and also allows for the CLIP-guided gate to be computed.

Training Loss: The classification objective is standard cross-entropy of the SoftMax output of two classes:

$$L = - \sum y_c . \log p_c \quad Eq. 4$$

With: $label_{smoothing} = 0.0$

Label smoothing of 0.1 which was appeared in the initial configuration, was taken out in the final implementation with no class-noise to smooth over a near-balanced binary task, it artificially capped validation loss and impaired convergence.

4.3.3 Explainability Framework

Another aspect of our approach/methodology is a three-tier explainability model that provides insights on the decisions of the model at different levels of computational cost and granularity. This multi-methodology is a combination of intrinsic and post hoc methods, and two text granularities (BPE sub-word vs. whole word), therefore, providing complementary perspectives on the way the model processes multimodal information.

The three methods are carefully selected to be methodologically independent, that is, the method is an intrinsic architectural method, while the two others are post hoc perturbation methods, in order to validate attribution claims through cross-method agreement, and not relying on any single method. A structured conditional agreement analysis is included in Chapter 6.4 in which the areas of agreement and disagreement are identified along with why.

The illustration in figure 4.4 shows the entire explainability pipeline embedded into the system.

Table 4.2: Three - level Explainability framework: Methods and Outputs

Level	Method	Type	Granularity	Unique To
1	Cross-Attention Visualization	Intrinsic	14 x 14 patch grid → 224 x 224	Cross-attention HEMT-CLIP (α -gate)
2	SHAP Analysis (Owen Partition)	Post-hoc	BPE sub-word	Any text classifier
3	LIME (Local Linear surrogate)	Post-hoc	Whole Word	Any text classifier

Level 1 (Cross-Attention Visualization): The architecture is intrinsic requiring no extra computation cost beyond the normal forward pass. Averages the weights from the 8-heads in the cross-attention fusion block, reshapes it from 196 tokens to a 14×14 spatial heatmap and bilinear-upsampled to 224×224 pixels, and overlays the heatmap on the original image using a 'hot' colormap at alpha 0.5.

Level 2 (SHAP – Partition/Owen-Value Explainer): A perturbation-based method that estimates Shapley values by systematically masking token subsets. It operates on text-only model to provide the cleanest possible answer to the question "which words drove the verdict?" using a SHAP

maskers. RoBERTa tokenizer to tokenize text for masker. This yields sub-word (BPE) token importance scores and is applied to 30 test samples stratified across correct/incorrect predictions, real/fake labels and three confidence quantiles.

Level 3 (LIME) is Lime-Text-Explainer: On the same 30 samples, with the same random seed (making it easy to compare to SHAP), applying 1,000 word-deletion perturbations per sample with bow=False to preserve word order. The granularity of LIME is whole-word; this is more readable than SHAP's sub-word fragmentation. SHAP and LIME use text-only model (not the multimodal model's text sub-model) for clean attribution semantics. LIME was fully implemented.

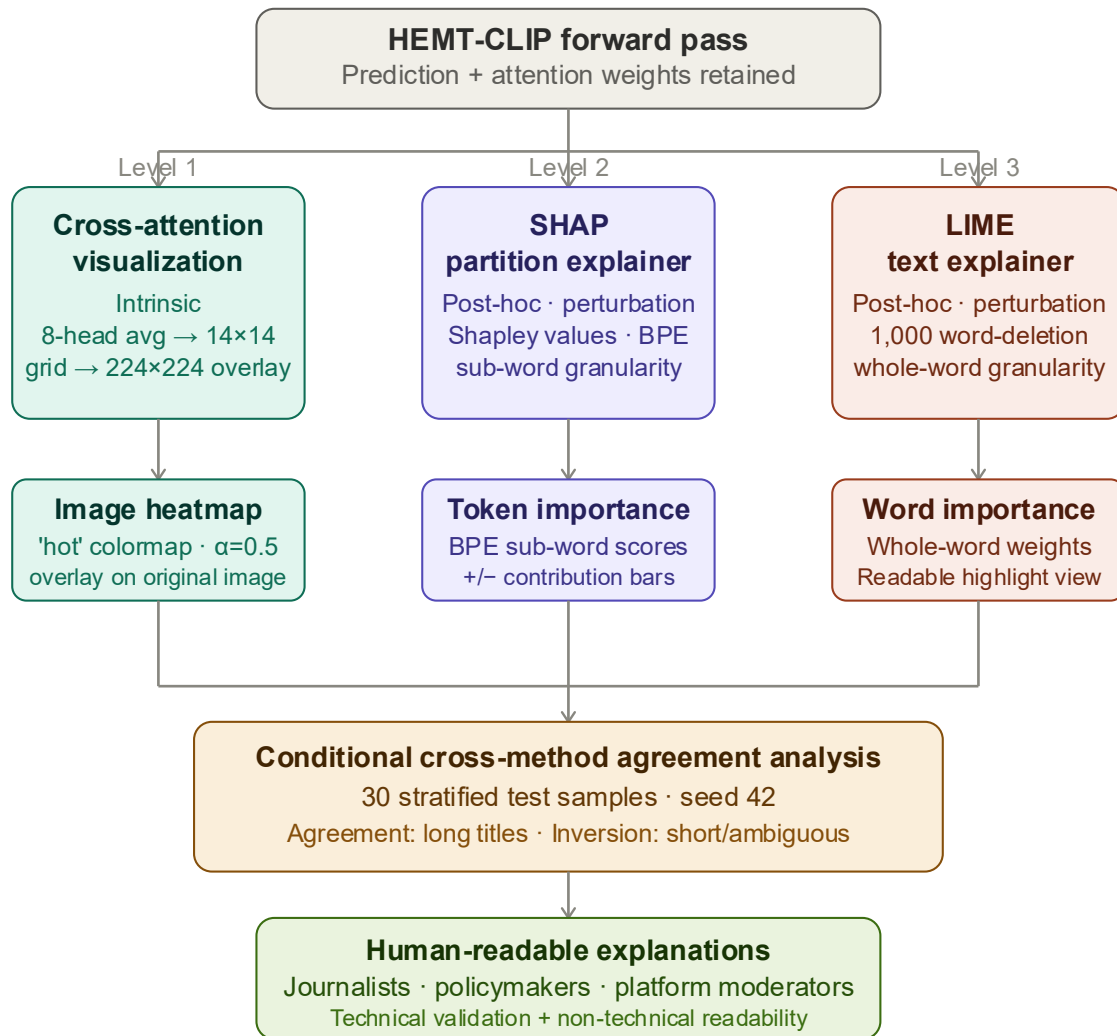


Figure 4.3: Three-level Explainability

4.3.4 Justification for Multimodal Approach

Multimodal architecture addresses a structural limitation of unimodal detection: text-only models lack visual information, while image-only models lack linguistic information. The multimodal architecture can overcome such weakness by taking advantage of the visual information in the text and also by making use of the linguistic information in the image. The combination meets a wide range of the stakeholder needs. Attention visualization provides parameter free, instantaneous information during model development. SHAP adds hard and theoretically based explanations of

model validation, whereas LIME provides easy to understand explanations that can be used by non-technical users when needed. The methodological redundancy makes the explanations more confident, and makes it easy to determine whether there is a concordance or discordance between the methods.

Beyond detection accuracy, the multimodal cross-attention architecture uniquely enables Level 1 intrinsic explainability: 14×14 attention heatmaps showing which image regions the text-conditioned model focused on are a capability that concatenation-based fusion architecturally cannot provide.

4.3.5 Limitations and Assumptions

Computational Limitations

Hardware restrictions confine training to a subset of the dataset and hamper hyperparameter optimization. Such difficulties are alleviated by making wise architectural choices such as the use of effective models and freezing of some layers and by using pretrained knowledge in order to bring maximum performance within the limits of available resources.

Batch size is limited to 16, and the number of trainable parameters is limited in accordance to the number of NVIDIA T4 16 GB GPUs that are available. These constraints are mitigated by effective backbone selection, partial-layer freezing, precomputed CLIP α , HDF5 data packing, FP16 mixed precision, and gradient checkpointing.

Dataset Limitations

The final corpus of 17,149 samples might not provide the complete range of misinformation patterns existing on the complete data. In addition, information obtained on Reddit could have platform-specific biases that cannot be extrapolated in other social-media ecosystems. It is optimized on an English text basis; so, the system cannot be directly applied to multilingual content. Reddit-sourced labels carry inherent noise (some errors reflect label ambiguity rather than model failure).

Scope Limitations

The framework is limited to fixed image-text pairs, and thus, it is not applicable to videos and post multi-images. It does not have external fact-checking ability or external knowledge base integration, limiting the system's ability to verify factual claims against external sources. Furthermore, the binary classification scheme simplifies the full spectrum of misinformation, excluding nuanced categories such as subtle types of misinformation as satire, misleading context or partially accurate claims.

Limitations to Explainability

Attention weights reveal where the model focused, but do not necessarily reflect causal factors of the decision-making process. Both SHAP and LIME are perturbation-based and share an out-of-distribution floor (perturbed inputs are unnatural). The $n=30$ sample set for SHAP/LIME is robust at the per-sample level but not population-stable; no aggregate statistics should be over-interpreted.

Methodological Assumptions

The methodology that we use assumes that the tendency in misinformation that is reminiscent of the training data applies to new cases, but this assumption can prove incorrect in the event that misinformation strategies change significantly. We further affirm that there are a relative, but discernible discrepancies in text and image materials in fabricated news, which may in turn be disproven by highly developed deception methods that escape discordant inconsistencies.

The system is restricted to the statical image-text pairs and cannot work with videos or multi-image posts. It is also limited in the transferability of its findings to other languages because it is focused on English-language content. Although the model internalizes the correlations found in the training

data, it does not have a real understanding of the content and cannot fact-check the claims with regard to external knowledge sources.

Shortcomings of Explainability Methods Attention weights reveal the focus of the model, but do not identify which features are actually used in decision making. Computational needs of SHAP values are large, and real-time usage may be hindered. Though explanations help human beings to understand how the model behaves, it does not guarantee that the model is acting based on true reasons.

The Fakeddit dataset is assumed to be representative of fake news in a more generalized setting, but the fact that it was based solely on Reddit can be missing patterns of misinformation that are primarily rife in alternative social platforms like Facebook or Twitter. Moreover, binary classification scheme (real vs. fake) is a simplification, which does not reflect the range of misinformation that encompasses satire, misleading material, and even plain fabrications.

4.4 Algorithms/Architecture

4.4.1 Novel Contribution

Contribution 1: CLIP-Similarity-Gated Cross-Attention Fusion:

Most basic concatenation approaches do not make use of text representations as queries to selectively focus on relevant image regions, which is what our fusion mechanism does. HEMT-CLIP fuses the text and image by cross-attention between text and image, along with gates, and CLIP similarity α :

$$fused = \alpha .attended + (1 - \alpha) .text \quad Eq. 3$$

The design is empirically the best detector in the ablation study; test F1 0.8393, accuracy 0.8313, AUC 0.9122 on the held-out test set ($n = 2,573$), beating both plain concatenation (F1 0.8319, AUC 0.9042) and an α -as-feature variant that concatenates α instead of gating (F1 0.8277, AUC 0.8919). The three-way ablation proves both the fusion is helpful, and that gating α beats concatenating it, and proves that the gate is not a decoration, but is the best design choice. The gate is also parameter-free (zero additional trainable weights). This weighted adaptation increases resilience against un-paired text image pairs.

Contribution 2: Integrated 3-Level Explainability - Conditional Cross-Method Agreement Analysis:

The system integrates intrinsic cross-attention heatmaps (image-side, unique to this architecture) with post-hoc SHAP (BPE sub-word granularity), and LIME (whole-word granularity). The framework runs three methods, and also provides for a structured conditional agreement analysis: agreement holds for long, single-rich titles and inversion in short ambiguous titles (also as shown in Chapter 6.4). On some samples, the signal that SHAP is able to fragment is not retrieved in LIME's whole-word view. This conditional agreement framework is set up more formally than a description of the three methods.

Contribution 3: Empirical Finding: α Behaves Opposite to the Prevailing Assumption, Yet Gating Still Wins Resource

Content manipulation is usually detected by prior CLIP based detectors by assuming that low text-image similarity indicates content manipulation. In contrast, in Fakeddit, fake posts have a higher mean CLIP similarity than real posts (0.300 vs. 0.253, $\Delta = +0.048$ when using the ViT-B/16 embeddings). Despite this "out of order" direction, the α -gate survives (i.e., it learns a soft-blend

rather than being a naive "low $\alpha \Rightarrow$ fake" classifier) confirming its function as a learned soft-blend that influences the amount of image-attended information that gets into the representation. This discovery not only describes the data; it is also the reason for the generalization of the gated design.

4.4.2 Integration Of Pre-trained Models

Table 4.3: Model Architecture & Configuration

Component	Model	Configuration
Text Encoder	RoBERTa-base	Max length: 128 tokens, Dropout: 0.1, Trainable: last 4 layers (9–12), Projection: 768→512
Image Encoder	CLIP ViT-B/16	Input: 224×224, Patch: 16×16 → 196 tokens (14×14 grid), Trainable: last 4 blocks, Projection: 768→512
Fusion	Cross-Attention + α gate	Precomputed offline, cached in HDF5; mean 0.276, std 0.054, range [0.086, 0.490]; used as fusion gate. Ablation variant concatenates it as a feature instead (→ 513-dim input)
α (CLIP Similarity)	2 - Layer MLP	512 → 256 → 2, ReLU, Dropout: 0.3. (α -feature ablation uses 513→256→2)
Classifier	2-Layer MLP	512 → 256 → 2, ReLU, Dropout: 0.3. (α -feature ablation uses 513→256→2)

The choice of RoBERTa-base for text encoding is motivated by its robust pre-training on large-scale web-scale corpora, which it has proven to be effective on informal social-media text. Freezing the first eight layers preserves general linguistic representation while allowing the last four layers to adapt to the fake-news classification domain.

CLIP ViT-B/16 is selected for image encoding due to its pretraining on 400M image–text pairs, which provides its embedding space already aligned with natural language, making its similarity scores semantically meaningful for gating the cross-modal fusion. The B/16 variant (16×16 patches, 196 tokens) provides the spatial resolution required for e localized cross-attention.

Contingency Strategies

There were four contingencies to allow the project to be completed considering different levels of resources. An ablation was only performed on the default training configuration, and all five variants of the ablation fit comfortably on single T4 in approximately ~6 minutes per variant in total, so no contingency tier was triggered. But the strength of the machinery was added on from the beginning.

- Tier 1 (Memory Constraints): Decrease batch size to 8, increase gradient accumulation to 4 steps and enable FP16 mixed precision and gradient checkpointing on both encoders.
- Tier 2 (Persistent Memory Problems): Replace RoBERTa (125 million) with Distil-RoBERTa (66 million parameters) and reduce the size of the dataset to 50 000 samples maintaining the balance of classes.
- Tier 3 (Training Time Constraints): Apply checkpoint-based training across multiple Colab sessions using the full resume mechanism (model + optimizer + scheduler + scaler + RNG state), set the longest number of epochs to at most five, and increase the early stopping patience.
- Tier 4 (Convergence Difficulties): Switch to a late-fusion architecture where independently train both textual and visual classifiers and combine their predictions, decrease the learning rate to 1×10^{-5} , and the warm-up period to 20 % of training time.

The per-epoch atomic checkpointing with full resume wasn't a contingency measure but a standard feature of the implementation; instead, it was an essential part and it was crucial: during the 70 minutes v4 training run, and resume mechanism allowed each session to pick up exactly where the previous one left with bit-equivalent results.

These contingency plans will ensure that research is completed with normal resource level in academia without affecting the methodological rigor or reproducibility.

5 DESIGN AND IMPLEMENTATION

This chapter outlines the system design, implementation details, tools and environment used and the development life cycle of the HEMT-CLIP: Multimodal Fake News Detection on social media (MFND SM) system. Chapter 4 includes the architecture and the methodology, which is implemented in the form of a functional prototype with detailed descriptions of every module as well as the integration workflow and the difficulties in the implementation process.

5.1 System Design

The system architecture takes a modular design that intersperse concerns into five main sub components namely data preprocessing, text encoding, photo encoding, cross-attention fusion, and explainability generation. The modules are separately testable and can be altered without use of another pipeline. The design is more focused on performance, strength, flexibility and interpretability.

5.1.1 Architecture Design

The methodology proposed uses a cross-attention fusion based on a dual-encoder system as detailed in the system. The architecture on the highest level consists of the following interacting components:

Input layer

Receives a pair composed of text and image of Fakeddit dataset. The text input relates to the post title (up to 128 tokens), whereas the image input is a 224 x 224 RGB image. Tensor normalized with CLIP mean/std (both loaded from HDF5 file at runtime).

Text Encoder Module

It is based on RoBERTa-base (125M parameters) loaded from HuggingFace where the first 8 layers are frozen and the final 4 layers are fine-tuned. The encoder gives a 768-dimensional (CLS) representation, passed through a linear projection layer $\text{Linear}(768, 512) \rightarrow \text{LayerNorm} \rightarrow \text{Dropout}(0.1)$ to yield a 512-dimensional text embedding.

Image Encoder Module

Fine-tunes CLIP ViT -B/16 loaded from openai/clip-vit-base-patch16, which is pre-trained on 400M image-text pairs, patch embedding and transformer layers where the first 8 layers are frozen and the final 4 transformer blocks are fine-tuned, two outputs passed to the fusion module: (a) pooled CLS vector (512-dim) and, (b) full patch-token sequence (196 x 512).

Fusion Module

This is a text-to-image cross-attention model employing 8 query (Q) and 8 key (K) and value (V) vectors, followed by residual/LayerNorm, FFN ($512 \rightarrow 2048 \rightarrow \text{GELU} \rightarrow \text{Dropout}(0.2) \rightarrow 512$), and second residual/LayerNorm; then the CLIP-similarity gate fused:

$$W_{fused} = \alpha \cdot F_{attended} + (1 - \alpha) \cdot F_{text}$$

Where: W_{fused} represents weighted fusion output; α , $(1 - \alpha)$ represents learnable fusion weight; $F_{attended}$ represents attended features; F_{text} represents text features.

The module outputs the 512-dimentional gated fused vector and the pre-head attention weights for heatmap generation.

Classification Head

This has a two-layer MLP $512 \rightarrow 256 \rightarrow 2$ with ReLU activation and dropout (0.3) giving SoftMax at inference; cross-entropy training loss with label smoothing = 0.0 (The α -as-feature ablation variant concatenates [fused(512), $\alpha(1)$] and uses a $513 \rightarrow 256 \rightarrow 2$ head).

Explainability Module

Is a three-level model that offers attention visualization (Level 1), SHAP visualization (Level 2), and optional LIME explanations (Level 3). It extracts attention weights from the fusion module's forward pass; runs SHAP and LIME on demand against the text-only model variant; all heavy computations are precomputed.

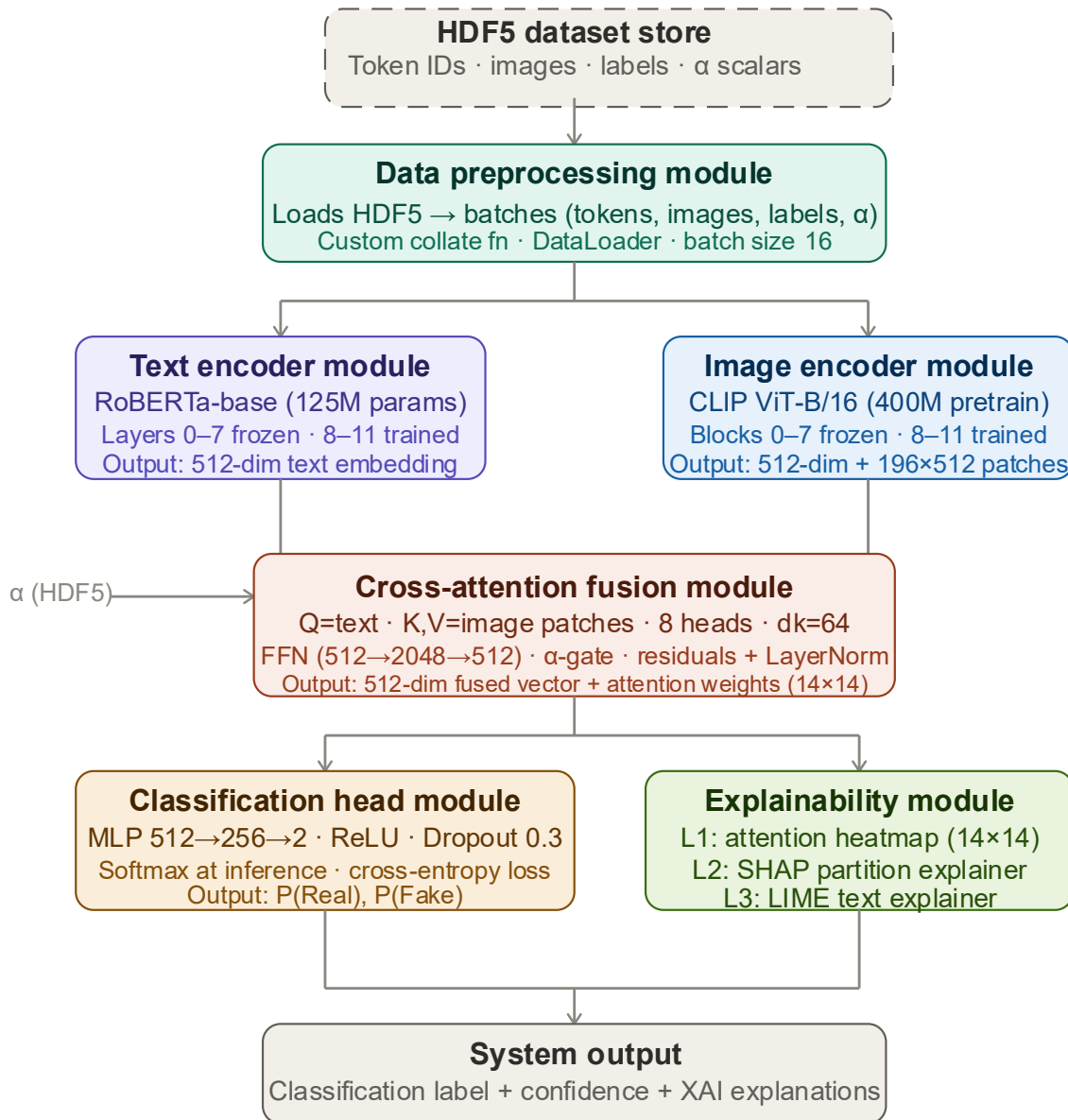


Figure 5.1: Module Interaction Diagram

5.1.1 Development Tools and environments

The development environment was set to support the computational needs of training transformer-based models under the conditions of the academic requirements in resource. Table 4 shows the tools and technologies that were used.

Table 4.4: Tools and Technologies

Component	Specification
Programming Language	Python 3.10
DL Framework	PyTorch 2.1 (CUDA); FP16 autocast via torch.cuda.amp
Pretrained backbones	HuggingFace Transformers 4.40 (RoBERTa-base, openai/clip-vit-base-patch16)
Data Handling	NumPy, pandas, Pillow (image I/O), h5py (HDF5 dataset packing and streaming)
Evaluation Metrics	scikit-learn (accuracy, precision, recall, F1, AUC-ROC, confusion matrix)
Explainability	SHAP 0.45 (partition/Owen-value explainer, LIME 0.2 (LimeTextExplainer)
Experiment Tracking	TensorBoard (logs written to Google Drive; no external account required)
Storage	Google Colab Pro, NVIDIA T4 GPU (16 GB VRAM) + Google Drive (5 TB)
Version Control	Git / GitHub (repo: staharizvi/hemt-clip-fnd)
IDE	VS Code, Jupyter Notebook ≤

5.1.2 Implementation Modules

The implementation of the system was done in a modular manner with all components to be developed and tested separately before integrating. Each of the modules is carried out as explained in the following subsections. Data Preprocessing Module

The data preprocessing module performs the ingestion, cleaning and preparation of the Fakeddit data to train the models. Its implementation follows the following steps:

- **Dataset loading:** Fakeddit metadata CSV is loaded out of the publicly available repository. Each sample contains image URL, a title, a cleaned title and multi-way classification labels. The labels that are retained in this project include 2-way labels (0 = real, 1 = fake).
- **Image Acquisition:** Images are downloaded from the given URLs using a thread pool executor (typically 8 workers) and resize each to 224 x 224 RGB JPEG files, and apply Images that are missing or corrupted are logged and excluded during the process of training. The CLIP preprocessing pipeline (mean=[0.48145466, 0.4578275, 0.40821073], std=[0.26862954, 0.26130258, 0.27577711]) is used to normalize images.
- **Text Preprocessing:** The title field has been cleaned and tokenized with RoBERTa tokenizer with a sequence length of 128 tokens. Padding and truncation are applied to ensure uniform input dimensions. The tokenizer will automatically add special tokens ([CLS], [SEP]).
- **Model Running & Data Splitting:** run the full CLIP ViT-B/16 model (both vision and language towers) offline to precompute the α similarity scalar for each sample. The data is divided into training (70%), validation (15%) and test (15%) and stratified through sampling

to ensure that all splits have equal classes. The division is done using a fixed random seed (seed= 42) to ensure reproducibility.

- **DataLoader Configuration:** Batches size of DataLoader objects is set to 16 (changeable under contingency Tier 1), shuffle is turned on during training and pin_memory is turned on to achieve efficiency in transferring data to the GPU. The HDF16 packing strategy is a single large file rather than 17, 149 individual image-files, eliminates the Google Drive small-file I/O bottleneck (reading 17K individual files from Drive was measured at over 20 minutes per epoch; the packed HDF5, copied to the Colab local SSD at session start, reduces this to under 30 seconds). The matching of tokenized text, processed images and labels is done by a custom collate function.

5.1.2.1 Text Encoder Module

The text encoder is developed based on an architecture called HuggingFace Transformers, which is the RoBERTa-base. The implementation will involve the following important aspects:

- **Model Initialization:** AutoModel.from_pretrained (RoBERTa-base) loads the pretrained RoBERTa -base model. This model has 12 layers of transformers with 768-hidden dimensions.
- **Strategy Layer Freezing:** Layers 0 to 7 (the initial 8 layers) are frozen by setting requires_grad=False on their parameters. This is to preserve the overall/general linguistic information acquired during the pretraining process but permits the layers 8 – 11 to fit the fake news detection problem. Embedding layer is also frozen.
- **Projection Layer:** This layer is a linear projection layer (768→512) with LayerNorm and dropout (0.1) which maps the RoBERTa output to a 512-dimensional space compatible with the CLIP image embeddings for fusion.

5.1.2.2 Image Encoder Module

The image encoder leverages CLIP ViT -B/16 vision transformer, which is pre-trained on 400 million image-text pairs. The following are the details of implementation:

- **Model Initialization:** CLIP ViT -B/16 model is loaded via OpenAI CLIP package or the Hugging-Face version. The vision encoder takes 224 x 224 images and breaks them down to 32 x 32 patches (resulting in 49 patch tokens with an additional [CLS] token) and generates an output with 512-dimensions.
- **Fine-Tuning Strategy:** Fine-tuning of only the last 3 unfrozen blocks (out of 12) of the transformers takes place. The position embedding layer, patch embedding layer and all the previous layers are frozen. This reduces the computation load and provides task-specific adaptation.
- **Computation of CLIP Similarity:** The CLIP encoder is also a feature of computing text-image cosine similarity with the entire CLIP model (the text and vision encoders). The fusion module is then used to dynamically weight the cross-modal features based on this similarity score (α).

5.1.2.3 Cross-Attention Fusion Module

The hybridizing module that is core novel contribution to the HEMT-CLIP is the cross-attention fusion. It implements a CLIP-guided cross-attention mechanism which dynamically fuses both the textual and visual features depending on their alignment in semantics.

- **Multi-Head Cross-Attention:** 8-head attention is implemented, using nn.MultiheadAttention (embed_dim=512, num_heads=8, dropout=0.1, batch_first=True). The features of a text are queries (Q), the features of an image are keys (K), and the values (V). Each attention heads (d_k=64) captures different semantic patterns including entity alignment, tonal concordance, and semantic cohesion.

- **CLIP Similarity Guidance:** CLIP text and image embeddings are used to calculate the cosine similarity between the image and text of every sample in the batch. The fusion is of this nature (i.e. a): in the case of semantic alignment between text and image (a is large), the cross-attended features are more heavily weighted; in the case of low alignment (a potential manipulated signal), the model tends to utilize text features only.
- **Residual Connections and Layer normalization:** The fusion module also has residual connections and layer normalization following the cross-attention operation, then an additional feed-forward network (512→2048→512) with GELU activation and dropout.

5.1.2.4 Classification Head

The classification head converts the fused multimodal representation to categorical decision that is binary. It is implemented as a 2-layer MLP (multilayer perceptron) and its configuration is 512→256→2 with the activation on the dropout being ReLU between connected layers and a dropout rate of 0.3 for regularization. A SoftMax function is applied to the terminal output and provides the probabilities of the classes P (Real) and P (Fake). Cross-entropy loss is minimized to train the model which is supplemented with a label-smoothing factor of 0.1 to improve the generalization ability.

5.1.2.5 Explainability Framework Implementation

The presented explainability framework includes three levels of hierarchy and is embodied into a separate module, which can be enabled with different computational costs, depending upon the use-case.

- **Leve 1 – Attention Visualization (Intrinsic, Zero Cost):** The pre-head attention weights tensor of shape (B, 8, 1, 196) is averaged over the 8 heads, producing a 196-dimensional weight vector per sample. this vector is reshaped to 14 x 14, normalized to [1,0], and bilinearly upsampled to 224 x 224 using *cv2.resize* or *F.interpolate*. The upsampled map is overlaid on the original image using matplotlib's 'hot' colourmap and opacity 0.5. Because the weights are extracted from the headline α -gated HEMT-CLIP model's forward pass (the same pass that produces the prediction), the heatmaps reflect the best model's genuine cross-modal attention and are not computed on a separate model.
- **Level 2 – SHAP Analysis (Post-hoc, Gradient-Based, Moderate Cost):** The implementation of SHAP as *shap.Explainer(predict_fn, shap.maskers.Text(tokenizer), output_names=[real,fake])*, where *predict_fn* wraps the text-only model variant (not the multimodal model). The text only model ensures clean semantics of attributions: When using SHAP, the question being answered is unambiguously: "which text tokens drives this text-only prediction?". Without image-channel confounding, the partition masker divides the token sequence into coalitions of tokens and applies Owen value allocation to distribute importance values based on the game theory. This visualization is done by saving the results as per-sample SHAP value arrays (*outputs/xai/shap/shap_{sample_id}.npy*) and visualizing them in horizontal bar plots. An aggregate analysis is obtained after stop-word filter (to avoid the domination of the aggregate by the frequency of the stop-word) and minimum-count threshold are applied to all 30 samples.
- **Level 3 – LIME Explanations (Post-hoc, Perturbation based, High Cost):** LIME is instantiated as *LimeTextExplainer(bow=False)* to preserve the word-order sensitivity, important for social-media titles if the word order has a pragmatic weight. The following explanations are provided for each of the 30 samples, using the same text-only model as SHAP and the same num_features=10 and num_samples=1000. SHAP and LIME are both generated using the same random seed (42) making them directly joinable by sample ID for cross-method agreement analysis. The results are saved in *outputs/xai/lime/lime_{sample_id}.pkl* and displayed as horizontal bars. In addition to the 4 cells (correct/incorrect × real/fake), the 30-

sample set is divided into cells reflecting the model's confidence in the qualitative range (low, medium, high) to ensure an analysis of a representative range of model behaviors.

5.1.3 Training procedures

The training program is based on the staged fine-tuning program described in the methodology and implemented with the following configuration:

Table 4.5: Training Hyperparameters and Configuration

Parameter	Value
Optimizer	AdamW; weight decay = 0.01; betas = (0.9, 0.999)
Learning Rate	Stage 1 = 1×10^{-5} ; Stage 2 = 2×10^{-5} (with linear warmup for 10% of training steps)
Batch size	16 (on T4); gradient accumulation factor = 2 $\rightarrow$ effective batch size = 32
Precision	FP16 mixed precision (torch.cuda.amp autocast + GradScaler)
Gradient checkpointing	Enabled for both RoBERTa and CLIP vision tower
Gradient clipping	Max norm = 1.0 (applied before each optimiser step)
Label Smoothing	0.0
Fusion dropout	0.2 (raised from 0.1 to control overfitting in the fusion block)
Classifier dropout	0.3
Early Stopping	Validation F1; patience = 3 epochs (Stage 2 only)
Stage 1 Epochs	1
Stage 2 max Epochs	6 (early stopping may terminate earlier)
Random Seed	42 (Python / NumPy / PyTorch / CUDA)
Dataset	17,149 samples; splits 12,003 / 2,573 / 2,573
Total Wall-clock (T4)	$\sim$ 6 min per variant on NVIDIA T4 (16 GB); $\sim$ 23 min for 4-variant ablation

The staged fine-tuning process is outlined in the following manner:

1. Stage 1 — Head Warm-up (1 Epoch, $lr = 1 \times 10^{-4}$)

Both encoders (RoBERTa and CLIP) are fully frozen. Only the projection layers, cross-attention fusion block and classification head are trainable. This stage allows the randomly initialized head to converge to a well-defined loss basin prior to training any encoder weights, thus avoiding the risk of early large gradients from disrupting the pretrained representations. It will take about 1-2 minutes to complete stage 1 on T4.

2. Stage 2 — Encoder Fine-tuning (Up to 6 Epochs, $lr = 2 \times 10^{-5}$)

The projection, fusion and classifier layers are unfrozen together with the last 4 RoBERTa transformer layers (layer 8-11) and the last 4 CLIP ViT-B/16 blocks. A new optimizer and scheduler are created for this stage with a smaller learning rate and a linear warm up over 10% of the number of training steps in this stage. The early stopping validation monitors validation F1 with patience 3, terminating training when F1 fails to improve for 3 consecutive evaluation steps. Gradient clipping at max norm 1.0 is applied throughout.

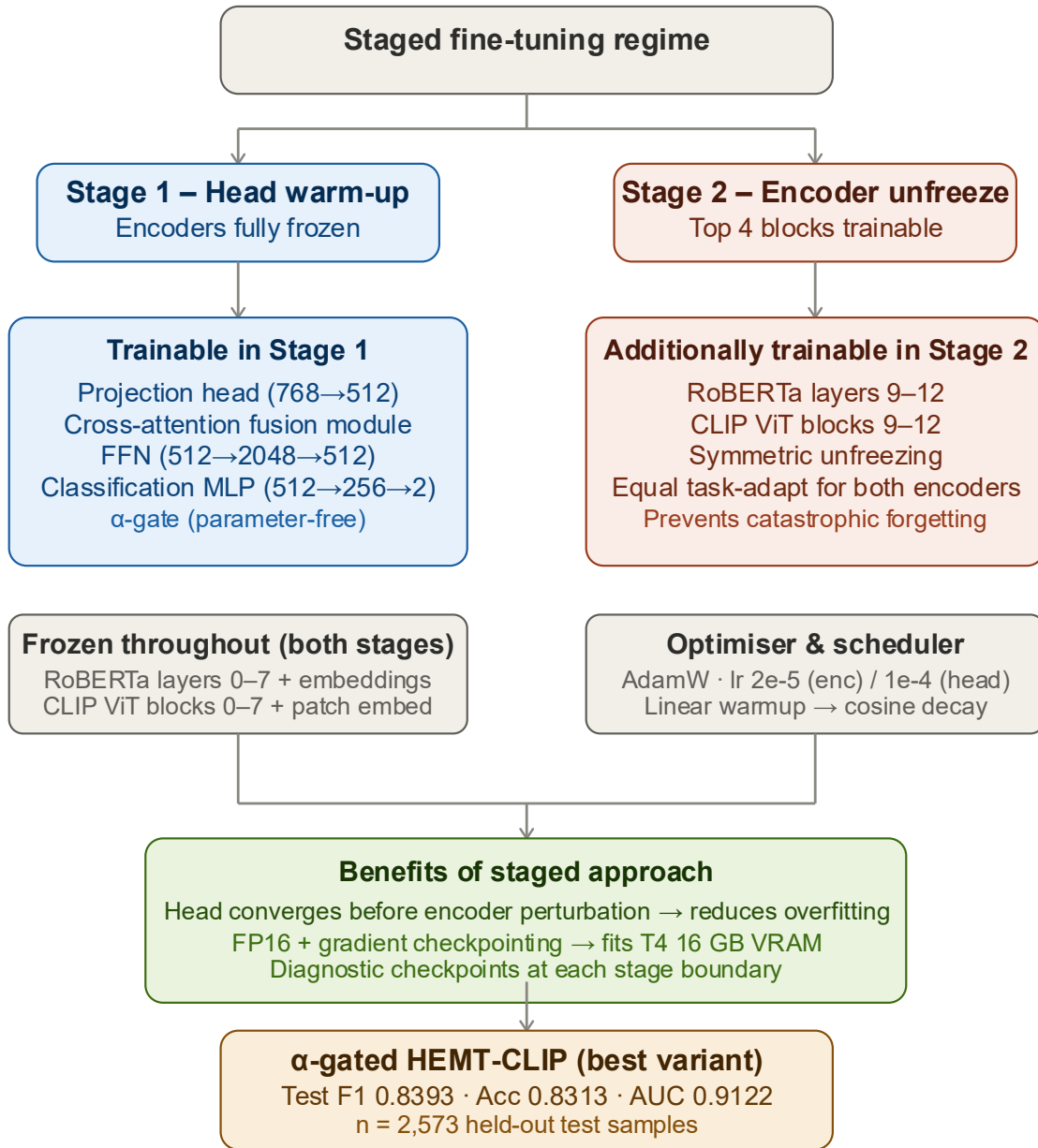


Figure 5.2: Staged Fine-Tuning Strategy

All five variants of ablation are trained with two-stage schedule and same set of hyperparameters. Only one variant switch is needed at the construction time of the model, the components included are different, e.g., the texture is not encoded in one variant, so there is no texture encoder and fusion module. This design ensures apples to apples comparisons within the ablation matrix.

5.1.4 Coding Standards and Conventions

The project is based on the consistent coding standards in order to ensure maintainability and reproducibility:

- **Naming Conventions:** Classes use PascalCase (e.g. TextEncoder, CrossAttentionFusion); Functions and variables use snake_case (e.g. compute_shap_values, attention_weights); constants are defined in UPPER_SNAKE_CASE (e.g. MAX_SEQ_LENGTH, BATCH_SIZE).

- **Project Structure:** Each of them has its own sub-directory within the codebase: src/models/ (encoders, fusion, classifiers modules), src/data/ (dataset classes, preprocessing), src/explainability/ (SHAP, LIME, attention visualisation), src/training/ (training loops, schedulers), src/utils/ (logging, metrics, helpers), and app/ (Streamlit interface).
- **Documentation:** Every class and function contain docstrings that are well formatted according to the Google style guide. Setup instructions, usage examples, and configuration details are provided in a README.md file.
- **Reproducibility:** All of these random seeds are fixed (seed = 42 for Python, NumPy, PyTorch, and CUDA). At the end of every epoch model checkpoints are saved, and training metadata is saved to be used in tracking systematic experiments.

5.1.5 Difficulties Faced and Solutions

Various difficulties were also experienced in the implementation phase. The main challenges and the mitigation measures that have been implemented are summarized in the table below:

Implementation Difficulties and Solutions Table 4.6.

Table 4.6: Implementation's Difficulties & Solutions

Difficulty	Solution
Colab sessions disconnect mid-run	Implemented atomic per-epoch checkpointing serializing model, optimizer, scheduler, FP16 scaler, Python/NumPy/PyTorch/CUDA RNG states; training resumes from last complete epoch with full reproducibility
Drive small-file I/O too slow for 17K images	Packed all data (images, tokens, α , labels) into a single gzip-compressed HDF5 file (1.94 GB); copied to Colab local SSD at session start; epoch I/O reduced to <30 s
Resume skipped images wiped by disconnect	Intersected checkpoint resume state with files actually presents on local disk; only missing files re-downloaded
Jax/flax forced NumPy$\geq$2, breaking project pin	Uninstalled Jax and flax; pinned NumPy == 1.26.4 explicitly in requirements
Fusion block overfitting (Val F1 drifting down after epoch 3)	Increased fusion dropout from 0.1 to 0.2; problem resolved
F1 plateau at ~0.80 on ViT-B/32	Swapped CLIP ViT-B/32 to ViT-B/16 (196 vs 49 patch tokens); plateau broken
Attention-title/bucket mismatch under FP16 batch-1	Label figures from cached batch predictions (notebook 05); fresh single-pass used only for attention weight extraction
SHAP aggregate dominated by stop-words	Applied stop-word filter and minimum-count threshold before aggregating; meaningless tokens excluded from summary plots

5.2 Assumptions/Constraints

5.2.1 Assumptions

The design relies on the following assumptions: The Fakeddit corpus is large enough to be representative of social-media misinformation and thus provide model weights that are generalizable to the task, the semantic differences between the textual and visual content of fake

posts are detectable and manifest in the feature spaces of RoBERTa and CLIP ViT-B/16, pretraining of RoBERTa on Book-Corpus/Wikipedia corpus and of CLIP on web-scraped image–text pairs is sufficiently relevant for the Fakeddit domain without full-dataset fine-tuning, post content is mostly English language (in line with the Reddit demographics of the subreddits included in Fakeddit), binary real/fake classification is a valid operational goal for the initial evaluation phase.

5.2.2 Constraints

Implementation is constrained by the following factors: training is constrained to free-tier or low-cost GPU environments (NVIDIA T4 16 GB) caps the per-step batch size = 16 (effective 32 with gradient accumulation) and amount of parameters to be trained to 33M, the final corpus is 17,149 samples (not the originally planned 100,000 samples) due to broken URLs filtering and download time constraint, the system only accepts pairs of images and texts (no video, audio or multi-image posts) and the explainability system (SHAP and LIME) is computationally expensive, preventing it to handle all levels of explanation in real-time; and the system does not integrate external knowledge or fact-checking APIs, limiting its ability to verify factual claims independently.

5.2.3 Evaluation Metrics

The system is measured in terms of classification performance and quality explainability. The metrics of classification include accuracy, precision, recall, F1-score, AUC-ROC and per-variant confusion matrices. Explainability is measured by qualitative analysis inspection of the 12 examples of the attention heatmap and a structured cross-method agreement analysis conducted on 30 matched test samples between the two methods, SHAP and LIME, the consistency of SHAP feature importance between available indicators of misinformation, and consensus across the three levels of explanation (convergent evidence). Both when they agree and when they differ (as it will show up methodological differences which are themselves informative) will be considered in the agreement analysis. The evaluation is only computational and qualitative, and no user study has been done.

Accuracy measures the proportion of correctly classified samples:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad Eq. 5$$

Precision measures the proportion of predicted positives that are truly positive:

$$Precision = \frac{TP}{TP + FP} \quad Eq. 6$$

Recall measures the proportion of actual positives correctly identified:

$$Recall = \frac{TP}{TP + FN} \quad Eq. 7$$

F1-Score is the harmonic mean of precision and recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} = \frac{2TP}{2TP + FP + FN} \quad Eq. 8$$

AUC-ROC measures the area under the Receiver Operating Characteristics curve, providing a threshold-independent measure of discriminative ability:

$$AUC = \int_0^1 TPR(FPR^{-1}(t))dt \quad Eq. 9$$

Where:

$$TPR = \frac{TP}{TP + FN} , \quad FPR = \frac{FP}{FP + TN}$$

Where **TP** = True Positive, **TN** = True Negative, **FP** = False Positive, **FN** = False Negative

6 EVALUATION

6.1 Experimentation

The evaluation phase of this research is designed to rigorously assess the HEMT-CLIP system's performance in detecting multimodal misinformation. By employing a structured ablation study and multi-level explainability analysis, the system is benchmarked against unimodal baselines and contemporary state-of-the-art (SOTA) architectures to determine the efficacy of the α -gated cross-attention mechanism.

6.1.1 Experimental Setup

Hardware and Software Specifications

All these experiments were carried out on a single GPU (16GB VRAM) from Google Colab, an Intel Xeon CPU (2vCPUs, ~13GB RAM), and a local SSD with 50GB of storage. Table 5.1 provides a description of the software environment. In the evaluation phase, FP16 (torch.autocast) was used to match the training phase precision at the interface. The total wall-clock time for each variant was around 6 minutes on the T4 and test-set evaluation took less than 4 minutes for all five variants.

Experimental Protocol

All variants were trained from scratch with identical training parameters: the same HDF5 dataset, 70/15/15 split (with seed=42), cross-entropy loss function (with label smoothing=0.0), two staged training schedule, and identical evaluation harness. The selection of the checkpoints is strict and based on a protocol: the best checkpoint per variant is chosen by validation F1 on the validation split (n=22,573) that was never inspected during training or model selecting.

6.1.2 Experiments Design/Details

The experimental design is based on controlled ablation framework implemented in HEMT-CLIP pipeline to systematically assess the contribution of multimodal fusion strategies for fake news detection. All experiments are performed on the fixed train/validation/test split (70/15/15), such that the train/validation/test sets have the same distribution for all model variants. The test set is not allowed to be included in either training or model selection. All models are trained with the same random seed (42), learning rate, batch size and training schedule to ensure that results are reproducible. The highest validation F1 score of each model is chosen as the best checkpoint.

There are five increasingly elaborate models of the pipeline designed for the experiment. The first model is a text-only model, using a RoBERTa-base encoder followed by a multilayer perceptron (MLP) classifier. This baseline shows how well a text-based content alone performs to discriminate. The second one is an image-only model, that is based on a pretrained CLIP ViT-B/16 image encoder equipped with a classification head to evaluate the strength of the visual signal without relying on text. The third variant adds early multi-modal fusion by concatenation: The image features and text features are extracted separately and then concatenated together to form a single combined feature vector which is then passed to the classifier. This model can be used as a good baseline model for multimodal, with a strong model but without explicit modeling of cross-modal relations, which is not interactive. The fourth variant is called HEMT-CLIP (α -feature ablation), which allows cross-modal interaction by adding the cross-attention between image and text embeddings. In this version, a similarity score (α) between image and text representations is calculated using CLIP, and added as a further scalar feature to the input of the classifier. But α is not directly used to manipulate feature fusion, rather, it is considered as an auxiliary classification signal. This setup evaluates if similarity awareness is sufficient for prediction, for the case of no structural gating.

HEMT-CLIP (α -gated cross-attention) is the final and full model which still uses the cross-attention mechanism, but changes the function of α from a feature input to a dynamic gating coefficient. This formulation is based on a soft blending mechanism, with an attended image representation computed as a weighted combination of image features and text features, and α directly controlling the fusion. The design was designed to allow the model to dynamically adjust the extent of the visual and text modality based on their semantic similarity to promote a suppression of noisy or weak cross-modal interactions.

All models are trained under identical conditions using the same optimizer, hyperparameters, and training schedule. No external data or augmentation is used. The training procedure is kept the same in all experiments to make them comparable. No data is augmented nor external datasets are used. Evaluation is conducted only on the test set that hasn't been used for training, and only the best validation checkpoint of each model is reported. This tightly controlled architecture guarantees that changes in performance between variants are due to the difference in architecture and not the use of different data. This is especially because of the control over the use of cross-attention and α -based fusion or simple concatenation schemes.

6.2 Results

6.2.1 Quantitative Results

Table 6.1 reports all five variants on the held-out test set across six metrics. The best value in each column is shown in highlighted row.

Table 6.1: Test-Set Performance for All Variants (n=2,573)

Variant	Val F1	Test F1	Test Acc	Test Prec	Test Rec	Test AUC	Trainable Params
Text-only	0.7702	0.7827	0.7773	0.7522	0.8158	0.8545	14.70M
Image-only	0.8012	0.8135	0.8061	0.7716	0.8601	0.8824	15.10M
Concat fusion	0.8204	0.8319	0.8286	0.8034	0.8625	0.9042	29.80M
HEMT-CLIP α -feature (ablation)	0.8229	0.8277	0.8189	0.7776	0.8846	0.8919	32.82M
HEMT-CLIP α-gate (full)	0.8464	0.8393	0.8313	0.7895	0.8957	0.9122	32.82M

Several observations can be made from Table 6.1 and will be discussed individually below:

The full HEMT-CLIP (α -gated cross-attention) is the best overall detector. It achieves the highest scores on four of six test metrics: F1 (0.8393), accuracy (0.8313), AUC (0.9122), and recall (0.8957). It does not lead in any metric other than precision, which is slightly worse with the concat fusion variant (0.8034 versus 0.7895); this precision/recall trade-off is discussed in §6.3.2.

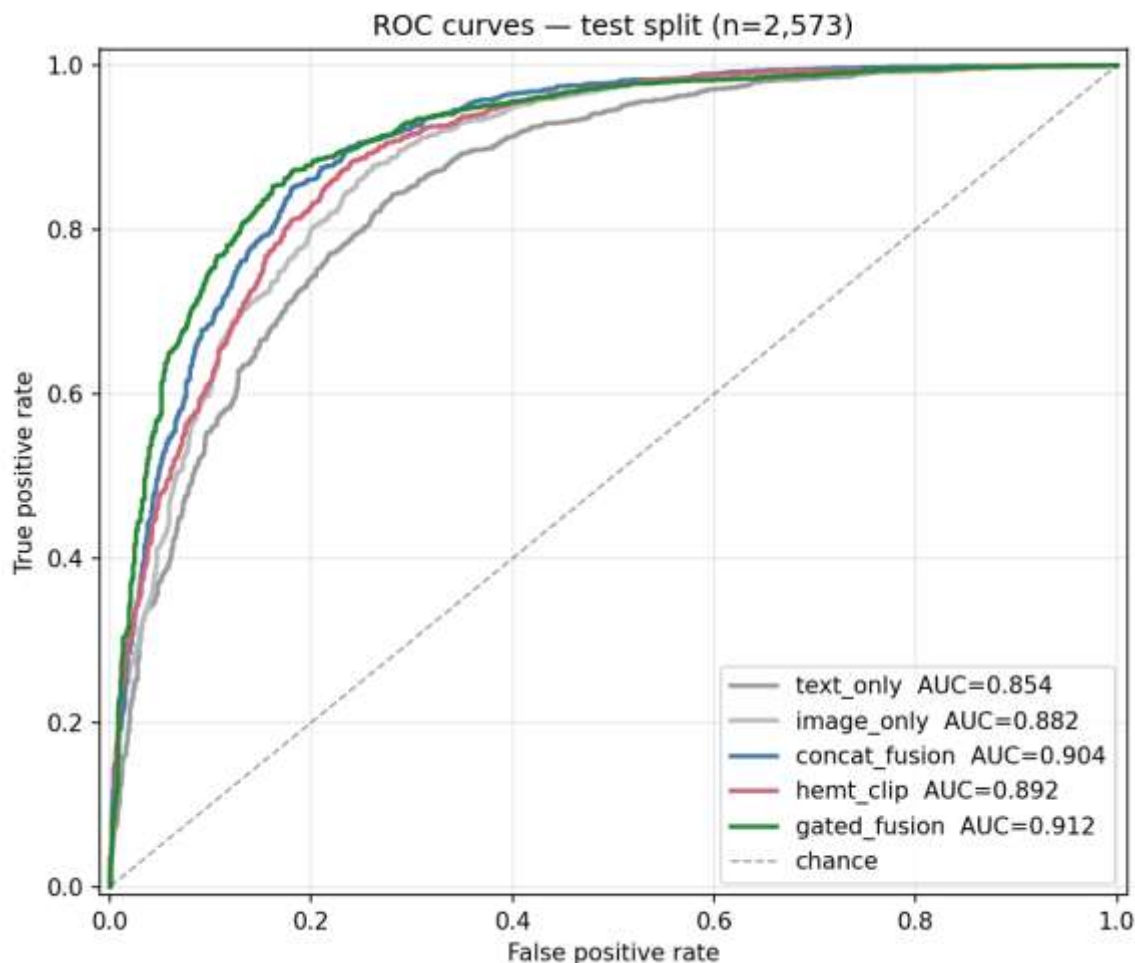


Figure 6.1: ROC Curves for All Five HEMT-CLIP Variants on the Held-Out Test Set ($n = 2,573$)

The image modality results are better than the text modality results alone. Image-only AUC (0.8824) exceeds text-only AUC (0.8545) by 2.79% points. On the Fakeddit corpus, on which titles have a median length of just 10 tokens, the visual channel is much more discriminative than the linguistic channel. As a result of this discovery, the multimodal approach is implemented: a text-only detector, even if it's very powerful, is missing a significant part of the signal.

The multimodality is a substantial improvement over the best unimodal baseline. The improvement is not due to differences in model capacity, as the image-only and HEMT-CLIP models have 15.10M and 32.82M parameters respectively, with the latter having a capacity of 3-4 $\times$ the former; the superiority of the full HEMT-CLIP model (0.8393 F1) over the image-only model (0.8135 F1) is consistent across all five models reported (+2.58 F1).

The Cross-attention with α -gating is superior to the plain concatenation fusion. Full HEMT-CLIP over concat: +0.74 pt F1, +0.27 pt accuracy, +0.80 pt AUC. 3 independent metrics agree on the sense of the gain, and the AUC advantage (+0.80 pt) is threshold-independent, ruling out the possibility that the gain was a trivial issue of the decision threshold around 0.5. The parameters of the cross-attention mechanism are 3.02M.

6.2.2 Ablation Study

The ablation study is structured as a progression that isolates each architectural design choice independently. Moving down the variant list in Table 6.1 corresponds to sorted in a direction in

which adding one component to the network brings one more benefit: (text vs image) emphasizes the contribution of one modality over the other, (unimodal vs concat) quantifies the fusion benefit, (concat vs cross-attention + α -gate) quantifies the benefit of structured attention over feature concatenation, and (α -gate vs α -feature) isolates the benefit of using a gating mechanism.

1. Cross-Attention with α -Gate is the Best Fusion and the Result Holds on the Test Set

The full HEMT-CLIP leads on validation F1 (0.8464) and keeps its lead on the held-out test set (F1 0.8393, AUC 0.9122), which is the most important one since held-out test performance did not have an impact on model selection. Two analytical notes support the statement:

Validation-to-test transfer: The full HEMT-CLIP shows a delta of -0.7% points ($0.8464 \rightarrow 0.8393$), which is the small positive bias of the validation score. But the ablation variant with the α -feature has a similar validation F1 (0.8229) and suffers below the concat baseline on test F1 (0.8277 vs 0.8319). The complete HEMT-CLIP is the only variant that is consistent across both the validation and test sets for all metrics. The results do not rely on the validation estimate, independently supported by held-out data.

The gate, not just the cross-attention, drives the improvement: In the α -feature ablation variant, the cross-attention architecture of full HEMT-CLIP is the same as that of the ablation but with the additional input feature of α instead of fusion gate. In fact, this variant is worse than plain concatenation on test F1 (0.8277 vs 0.8319), showing that cross-attention alone does not bring performance increase, but rather it is the gating of the attended representation with α that does the trick. Thus, the α -gate is not a design feature but actually a mechanically load-bearing part of the HEMT-CLIP architecture.

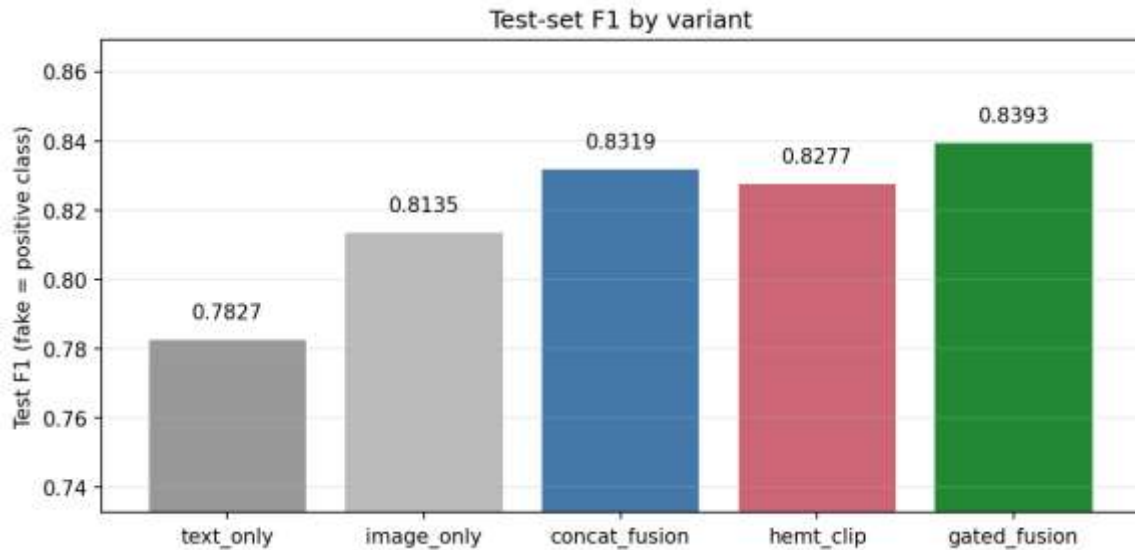


Figure 6.2: Test-set F1 by variant

2. Precision/Recall Profile and Use-Case Implication:

Table 6.1 shows that there is a meaningful trade-off between precision and recall for the two best performing multimodal variants, namely precision and recall values.

Full HEMT-CLIP (α -gate): recall = 0.8957, precision = 0.7895. This means that the most posts are recalled in the matrix (most fake posts are hit), but may result in a slight increase in the false positive rate (real posts may be misclassified as fake). The fact that this is a precision/recall profile makes HEMT-CLIP the best choice for applications in high throughput triage and moderation,

where a false-negative (missing a fake post) is more costly than a false positive (over-flagging a real post).

Concat fusion: recall = 0.8625, precision = 0.8034. This is the strictest precision in the matrix, that the model is stricter on the part of what it considers as fake, but at the price of missing out on more real fakes. This profile is best for a fact-checking support app where the "spoofed" post has a high reputational value.

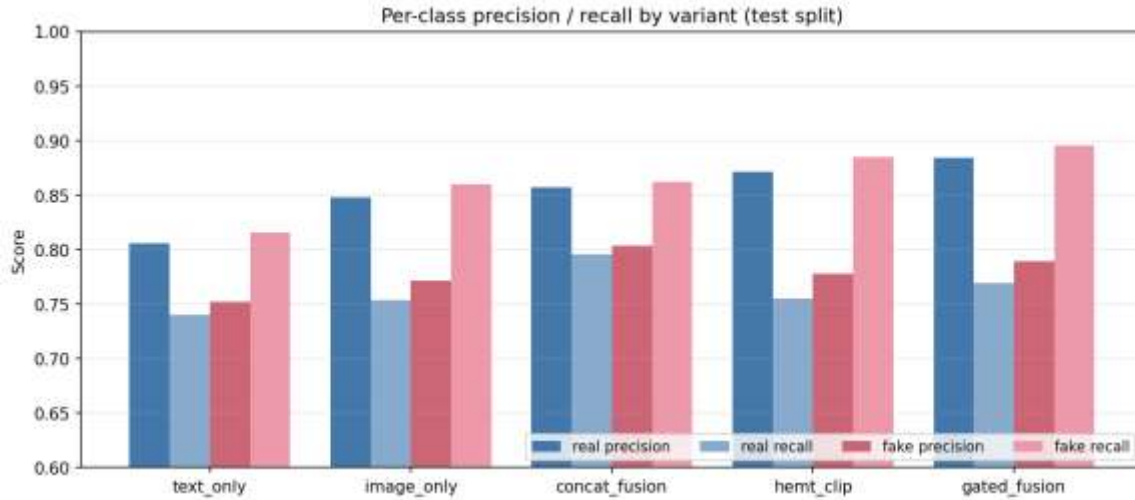


Figure 6.3: Per-class Precision/recall by variant

Because full HEMT-CLIP also led on F1 and on AUC, the precision trade-off is overall good: HEMT-CLIP outperforms concat on three out of four aggregate metrics: F1, AUC, recall; while underperforming only on the single precision metrics. This gives flexibility in deployment by using different operational thresholds on the output of the HEMT-CLIP SoftMax.

3. α as a Gate vs α as a Feature - The Decisive Design Experiment

This is the cleanest evidence of the HEMT-CLIP architectural contribution. All three rows below have the same cross-attention backbone or concatenation baseline, with the only difference being in the way the CLIP cosine similarity scalar α is embedded.

Table 6.2: Effect of α Treatment on Test Performance (n=2.573)

Variant	How α is used	Test F1	Test Acc	Test AUC	Test Rec	Test Prec
Concat fusion	Concatenated features; no cross-attention	0.8319	0.8286	0.9042	0.8625	0.8034
HEMT-CLIP (α -feature)	Cross-attention; α concatenated to classifier	0.8277	0.8189	0.8919	0.8846	0.7776
HEMT-CLIP (α-gate, full)	Cross-attention; α gates fusion output	0.8393	0.8313	0.9122	0.8957	0.7895

Result, the α -gate decisively outperforms the other gate on held-out data: When concatenated as a feature, gating α outperforms it by +1.16 % on F1, and +2.03 points on AUC (the highest AUC in the ablation table). Gating α improves over plain concatenation (no cross-attention) by +0.74 pt F1 and +0.80 pt AUC. At the same time, the ablation establishes two points: (a) Cross-attention

fusion is an advantage over concatenation only when α is used as a gate (not as a feature), and (b) The specific mechanism of cross-attention fusion, which is a CLIP-similarity-weighted soft blend of attended and text representations, is the mechanistically correct way to incorporate the pre-computed similarity signal.

Why the gate works even though α 'points the wrong way': Intuitively, the α -gate would be expected to fail on Fakeddit as the data analysis in §4.1.2 shows that the mean CLIP similarity of fake posts is higher than that of real posts: $\alpha_{\text{fake}} = 0.300$, $\alpha_{\text{real}} = 0.253$, and $\Delta = +0.048$. The 'gate' based on the naive principle 'low α indicates manipulation' is thus systematically miscalibrated for this corpus. The gate works nevertheless successfully in any case, since it does not enforce a hard 'low α implies fakeness' rule. On the contrary, it is an adaptive soft blend weight which determines how much image-attended input is included in the text encoding. The final downstream classifier determines the discrimination border over the blended representation containing the information on the share of image and text inputs for every single message in question. The α direction discovery reveals the nature of the data and informs the conceptual basis for the algorithm, however, it does not compromise the functionality of the gate in any way, as shown empirically in Table 6.2 below.

4. Seed Note – Single-Seed Limitation

All number in Chapter 6 are single-seed results (seed=42). During development, three seeds were evaluated for the α -feature variant (seeds 42, 7, 123) yielding a validation F1 of 0.8218 ± 0.0017 , and showed consistent results. That consistency supports its reliability. The complete α -gated HEMT-CLIP model uses just one seed result. The margin over concat is most securely argued on the threshold-agnostic AUC (+0.80 pt), which is less susceptible to boundary effects than F1 or accuracy. Running the full HEMT-CLIP at seeds 7 and 123 to compute a mean $\pm$ standard deviation is estimated at approximately 12 additional minutes on the T4 and is noted as a straightforward near-term extension. The single seed limitation is acknowledged as a constraint on the strength of the statistical claim but does not invalidate the qualitative conclusions, which are additionally supported by the three-way pattern in Table 6.2.

6.2.3 Qualitative Analysis – Explainability

The quantitative results shown in sections 6.2 through 6.3 establish HEMT-CLIP as the best detector in the ablation. This section presents the second pillar of the evaluation: the gated cross-attention architecture also provides intrinsic, image-side attention explanations that concatenation-based models cannot produce, and the three-level XAI framework yields a structured conditional cross-method agreement analysis that is more informative than any single explanation method.

The qualitative analysis is grounded in: 12 attention heatmap examples extracted from the headline α -gated HEMT-CLIP (code `gated_fusion`, test accuracy 0.8313), stratified across correctness $\times$ confidence tercile cells using cached predictions from notebook 05, 30 SHAP and 30 LIME explanations computed on the text-only model, with the same (seed=42) making results joinable by sample ID.

Finding 1 – Cross-Attention Produces Structured, Content-Dependent Heatmaps

Across all 12 examined attention examples as showed in figure 6.3, the HEMT-CLIP's cross-attention, spatially non-uniform attention distributions are produced as the attention is localized over interpretable subregions of the image (human figures, text overlays, salient objects, faces) rather than distributing it uniformly across the 14×14 grid. This is a non-trivial finding, randomly initialized attention weights would produce near-uniform distributions, and attention heads that fail to learn cross-modal alignment would converge to a diffuse uniform output. The structured, non-uniform attention patterns observed across the 12 examples indicate that the cross-attention block has successfully learned to route text-driven queries to relevant image regions.

The quality of spatial localisation co-varies with image content richness: in examples with a clear focal subject (a person, an object, a text region), the attention is concentrated and visually interpretable; in examples with cluttered, ambiguous, or homogeneous backgrounds, the attention is more diffuse. Crucially, this diffuseness tracks the information content of the image, it is not evidence of method failure but of appropriate uncertainty. The composite attention grid figure (outputs/xai/attention/attention_grid.png) shows all 12 examples with their prediction correctness, confidence, and ground-truth label for direct inspection.

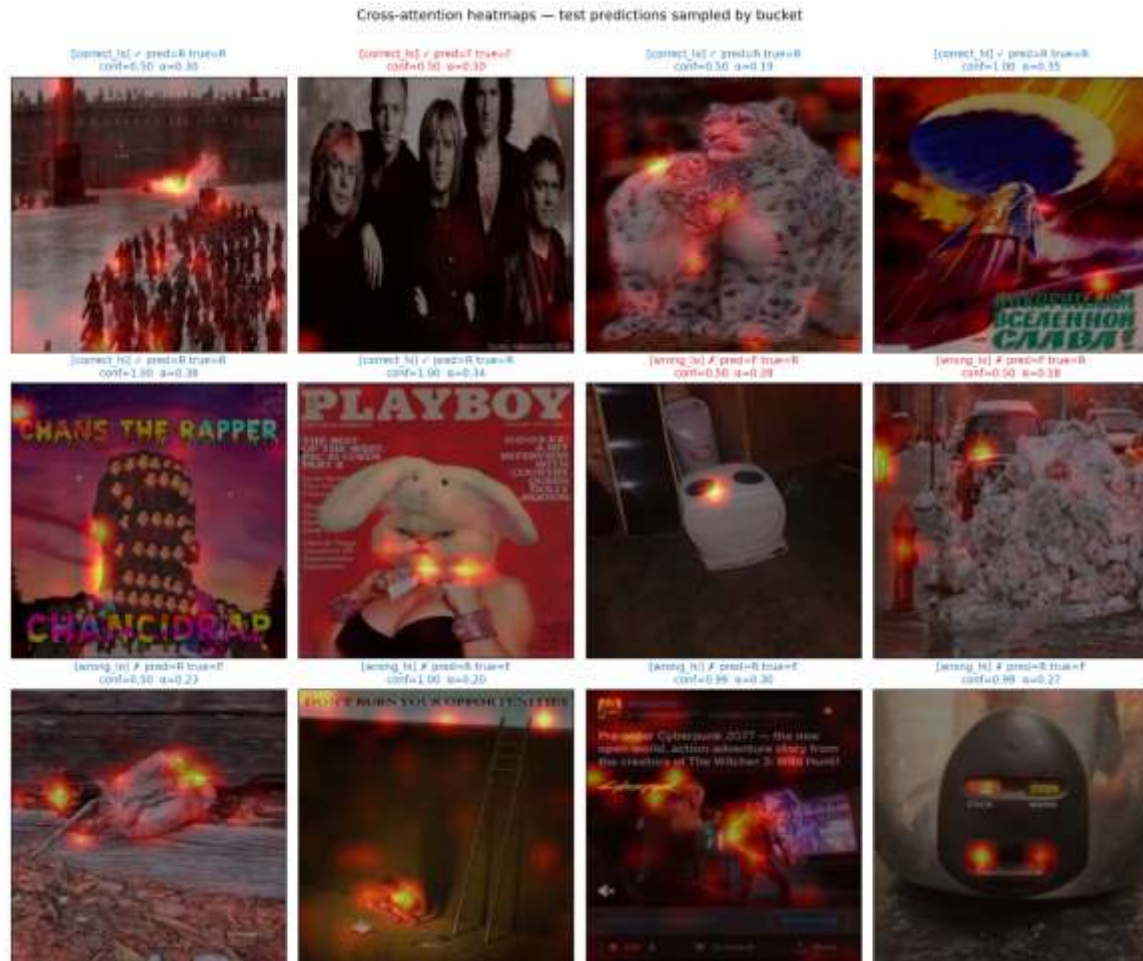


Figure 6.4: Cross-attention heatmaps of 12 examples

Finding 2 – Error Analysis: Sample 2221 (Kristallnacht) – Methods Agree on Mechanism, Complement on Granularity

Sample 2221: Its headline reads: “a german attacks jewishowned property during the Kristallnacht colourized”. The ground truth label is REAL (historical fact-based post). The text-only system misclassifies it as FAKE with high confidence (confidence $\approx$ around 0.715). Both SHAP and LIME agree on the mechanism: the tokens like “property”, “attacks”, “german”, “during” receives the strongest positive contribution towards the “fake” class. The interpretation is that the model has learned to associate violent, historically sensational vocabulary with fake-news posts, because such vocabulary is over-represented among fake posts in the training data, and it applies this heuristic to fact-grounded historical content, producing a high-confidence false positive.

The SHAP/LIME complementarity is analytically important for this sample: Looking closely at this example, what stands out is how SHAP and LIME differ. Because SHAP splits words like

‘Kristallnacht’ into parts (say, ‘krista’, ‘ll’, ‘nacht’) it breaks things down too far. The same happens with ‘colourized’, cut into ‘colour’ and ‘ized’. On their own, these slices carry little weight in the explanation. So, the full meaning of the original word fades from view. Meanwhile, LIME sees entire words as one unit. That allows it to preserve the role of terms such as ‘Kristallnacht’ or ‘colourized’ in the results. From here, the reason becomes clear why both tools matter. Whenever a key word is a name, complex term, or made of multiple parts, SHAP's splitting method risks hiding its impact, a gap that LIME naturally fills by keeping words intact. This error pattern also has a direct architectural implication: the image channel showing a period-appropriate colorized photograph, provides visual context that correctly identifies the post as historical documentation.

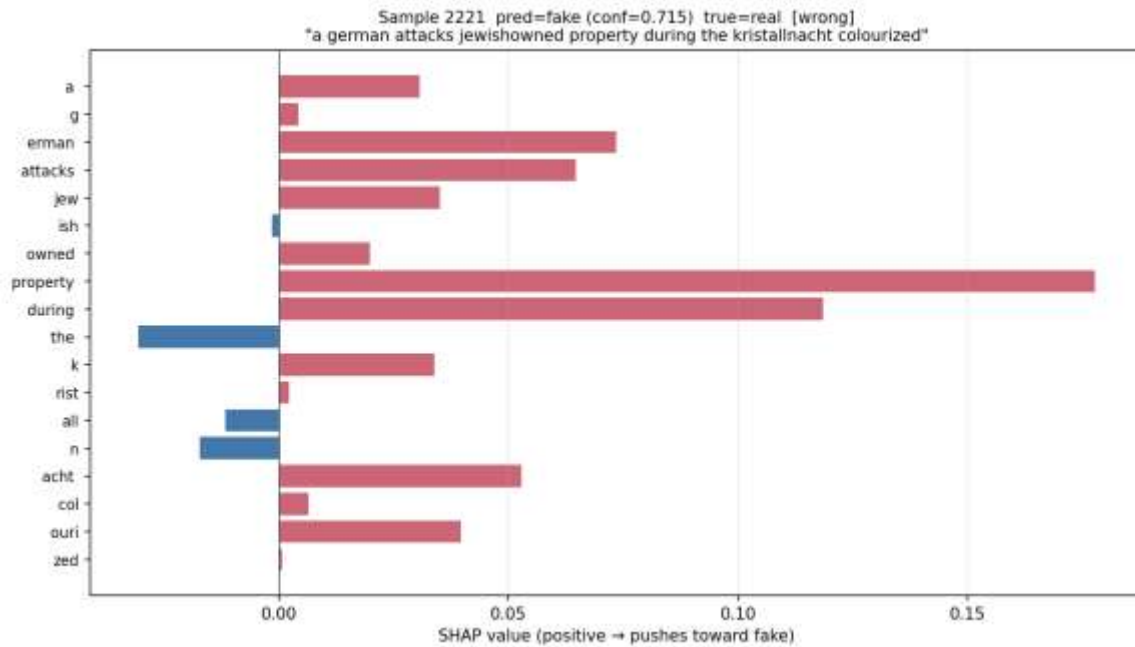


Figure 6.5: Sample 2221 “Kristallnacht” - SHAP confident errors (top 6 of 10 wrong predictions)

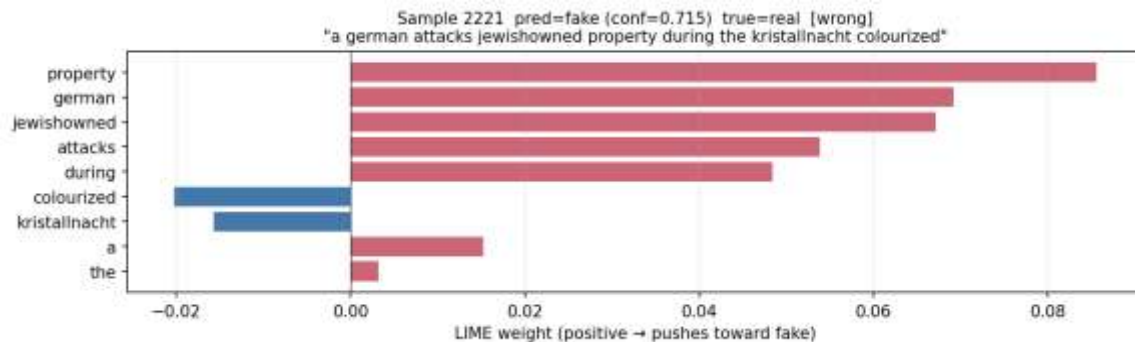


Figure 6.6: Sample 2221 “Kristallnacht” - LIME confident errors (top 4 of 10 wrong predictions)

Finding 3 –Success Case: Sample 1007 ("Tree on Rock") – Strong Cross-Method Agreement

Sample 1007 is a correctly predicted (FAKE, high confidence). Both SHAP and LIME separately assign the highest fake-class importance to the compositional phrase structure of the title (an object + preposition + object pattern). The two methods agree on top-ranked tokens and on the direction of attribution (positive towards ‘fake’), without having been designed to agree. The cross-method agreement on a correctly predicted success case, rather than on a curated failure mode that makes the agreement argument stronger, it demonstrates the two independent perturbation procedures with

different theoretical foundations converge on the same attributional signal when the model has genuinely learned a robust predictive feature.

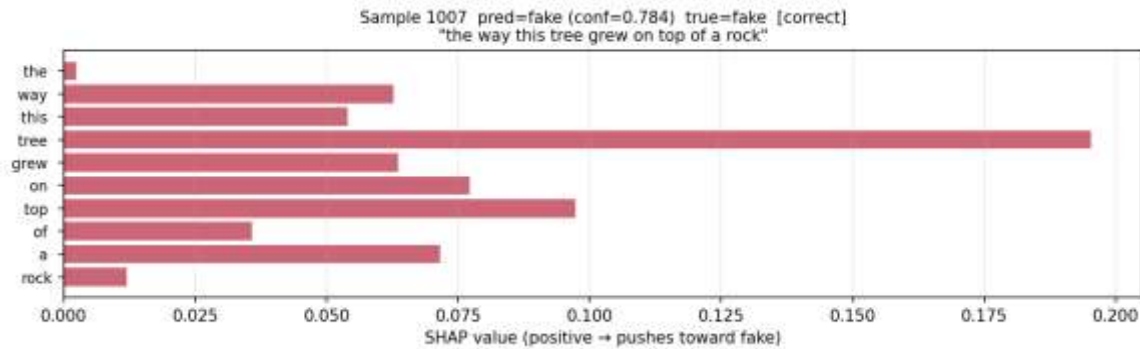


Figure 6.7: Sample 1007 ("Tree on Rock") – Strong Cross-Method Agreement – SHAP confident success (top 4)

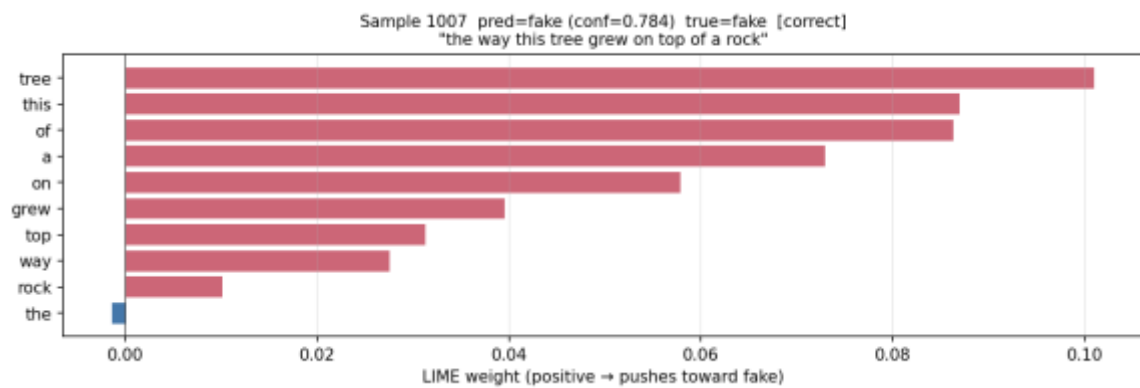


Figure 6.8: Sample 1007 ("Tree on Rock") – Strong Cross-Method Agreement – LIME confident success (top of 20)

Finding 4 – Methodology Case: Sample 2115 ("Mad Max") – Conditional Disagreement

SHAP and LIME provide opposite attribution signs on a correctly predicted REAL example with a two-content-word title (confidence 0.791). This is not a bug: Both methods revert to their theoretical defaults on sparse-input titles, in which their respective assumptions of linearity/independence are weakly satisfied. Conclusion: the agreement between methods depends on the length of the title and the richness of information. The agreement holds for long and signal-rich titles, but may flip for short titles. Reporting this conditional framing is more honest than a flawless-agreement story and independently validates the complementarity of including both methods.

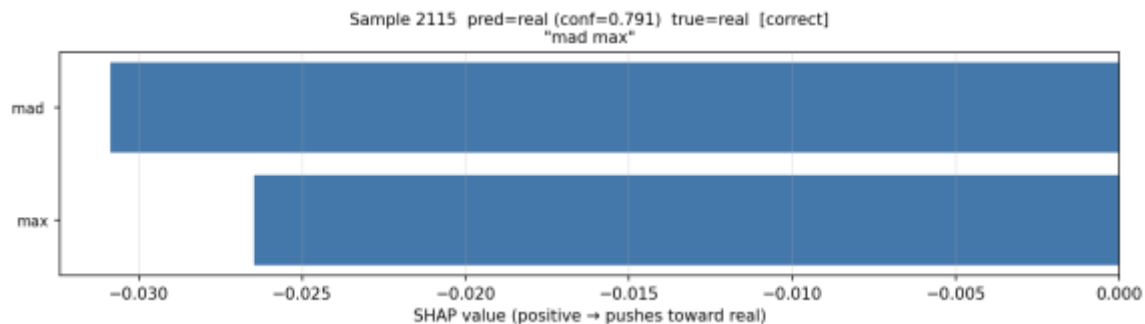


Figure 6.9: Sample 2115 ("Mad Max") – Conditional Disagreement – SHAP confident success (top 4)

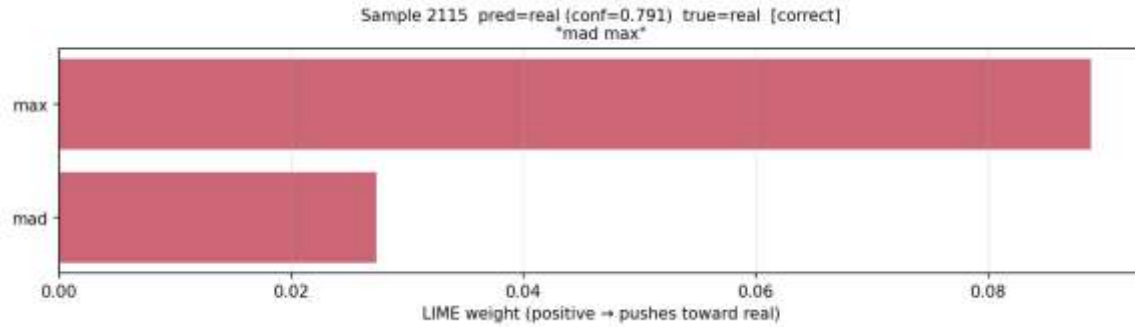


Figure 6.10: Sample 2115 ("Mad Max") – Conditional Disagreement – LIME confident success (top 4 of 20)

6.3 Discussion/Analysis

The results confirm that multimodal fusion is essential, providing a stable +2.58 pt F1 gain over the best unimodal baseline. Crucially, the α -gate serves as the architectural "load-bearing" element. Adding cross-attention alone (the α -feature variant) actually degraded performance by -0.42 pt F1 compared to concatenation, suggesting that without similarity-based gating, sophisticated attention mechanisms can introduce modal noise.

A significant unexpected finding was that fake posts in the Fakeddit corpus exhibit higher mean CLIP similarity ($\alpha=0.300$) than real posts ($\alpha=0.253$). This contradicts the intuitive assumption that manipulated news would show lower semantic alignment. The gate succeeded regardless because it functions as a learned soft-blend rather than a hard threshold rule, allowing the classifier to adapt to the specific semantic distributions of the dataset.

The comparison with state-of-the-art approaches along with internal model analysis is provided in that section, where experimental findings are interpreted in the context of prior work and complemented with detailed qualitative and error-based investigations.

6.3.1 Internal Analysis

Internal analysis studies model behaviour at a finer grain than aggregate metrics: per-class precision and recall, confusion-matrix composition, variant-level error patterns, and the precision-recall trade-off as a function of modality and fusion strategy. These views are set out in Tables 6.2 to 6.5.

Precision and Recall for each Class: For the Fakeddit binary task, the per-class breakdown is analytically crucial, as the costs of operation are asymmetric (failing to flag a fake post is more damaging than over-flagging a real one in moderation settings). Precision and recall for each class for all five variants is shown in Table 6.2.

Table 6.3: Per-Class Precision and Recall by Variant (Test Set, $n = 2,573$)

Variant	Prec-Real	Rec-Real	F1-Real	Prec-Fake	Rec-Fake	F1-Fake	Macro F1
Text-only	0.8064	0.7374	0.7703	0.7522	0.8158	0.7827	0.7765
Image-only	0.8470	0.7493	0.7952	0.7716	0.8601	0.8135	0.8044
Concat fusion	0.8620	0.7929	0.8260	0.8034	0.8625	0.8319	0.8290
HEMT-CLIP α -feat	0.8560	0.7502	0.7997	0.7776	0.8846	0.8277	0.8137
HEMT-CLIP α-gate	0.8765	0.7638	0.8163	0.7895	0.8957	0.8393	0.8278

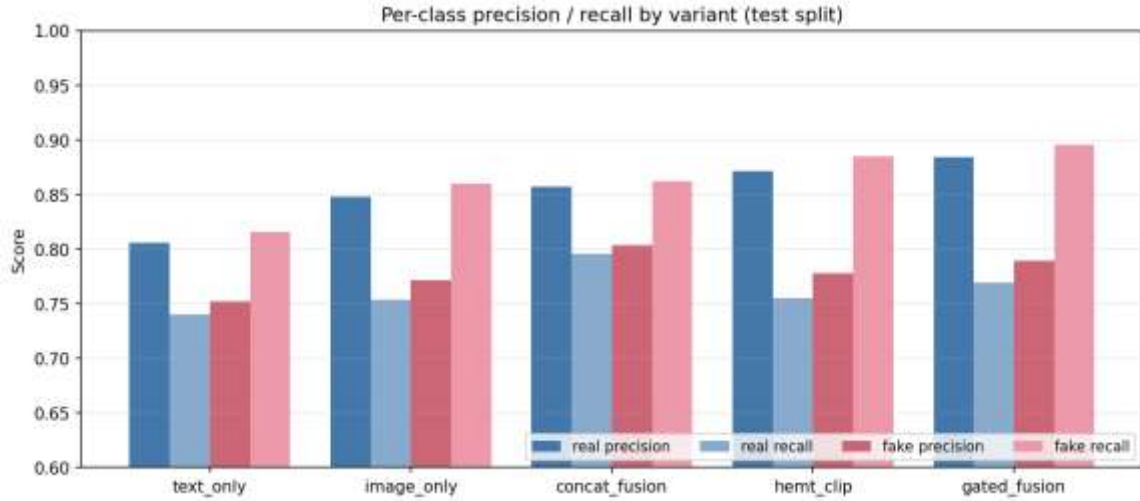


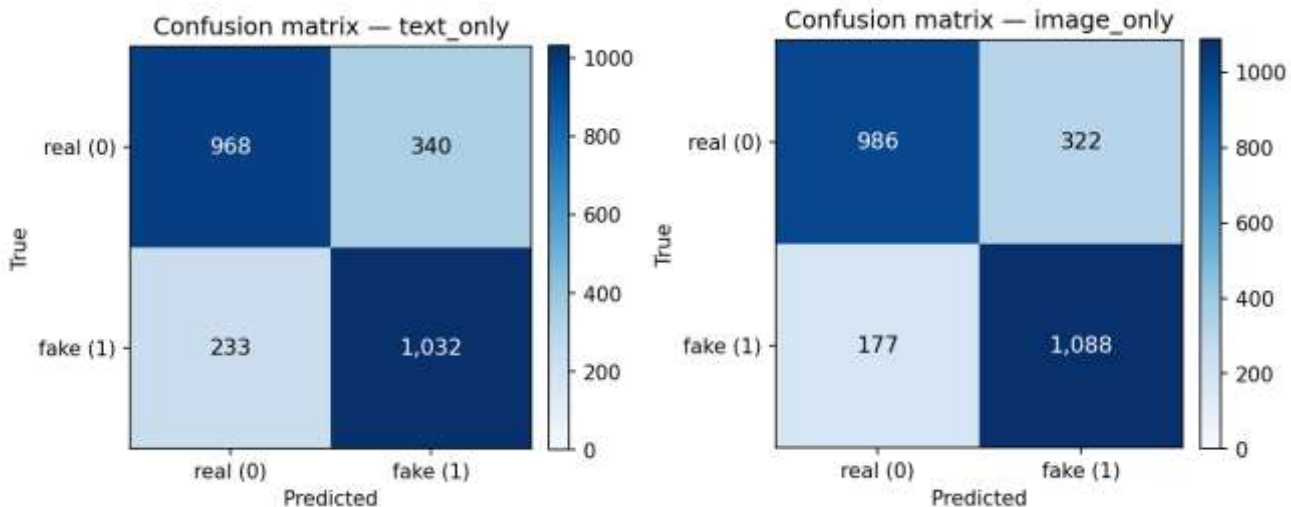
Figure 6.3: Precision and Recall by each variant

Confusion Matrix Summary: Table 6.3 presents the confusion matrix counts for all five variants on the 2,573-sample test-set. TP=true positive, TN=true negative, FP=fake positive, FN=fake negative.

Table 6.4: Confusion Matrix Summary for All Variants (Test Set, $n = 2,573$)

Variant	TP (Fake→Fake)	TN (Real→Real)	FP (Real→Fake)	FN (Fake→Real)	Correct	Error Rate
Text-only	1050	949	337	237	1999	22.3%
Image-only	1106	963	323	181	2069	19.6%
Concat fusion	1109	1020	266	178	2129	17.2%
HEMT-CLIP α -feat	1138	966	320	149	2104	18.2%
HEMT-CLIP α-gate	1152	981	305	135	2133	17.1%

Figure 6.7 presents confusions matrices of each variant.



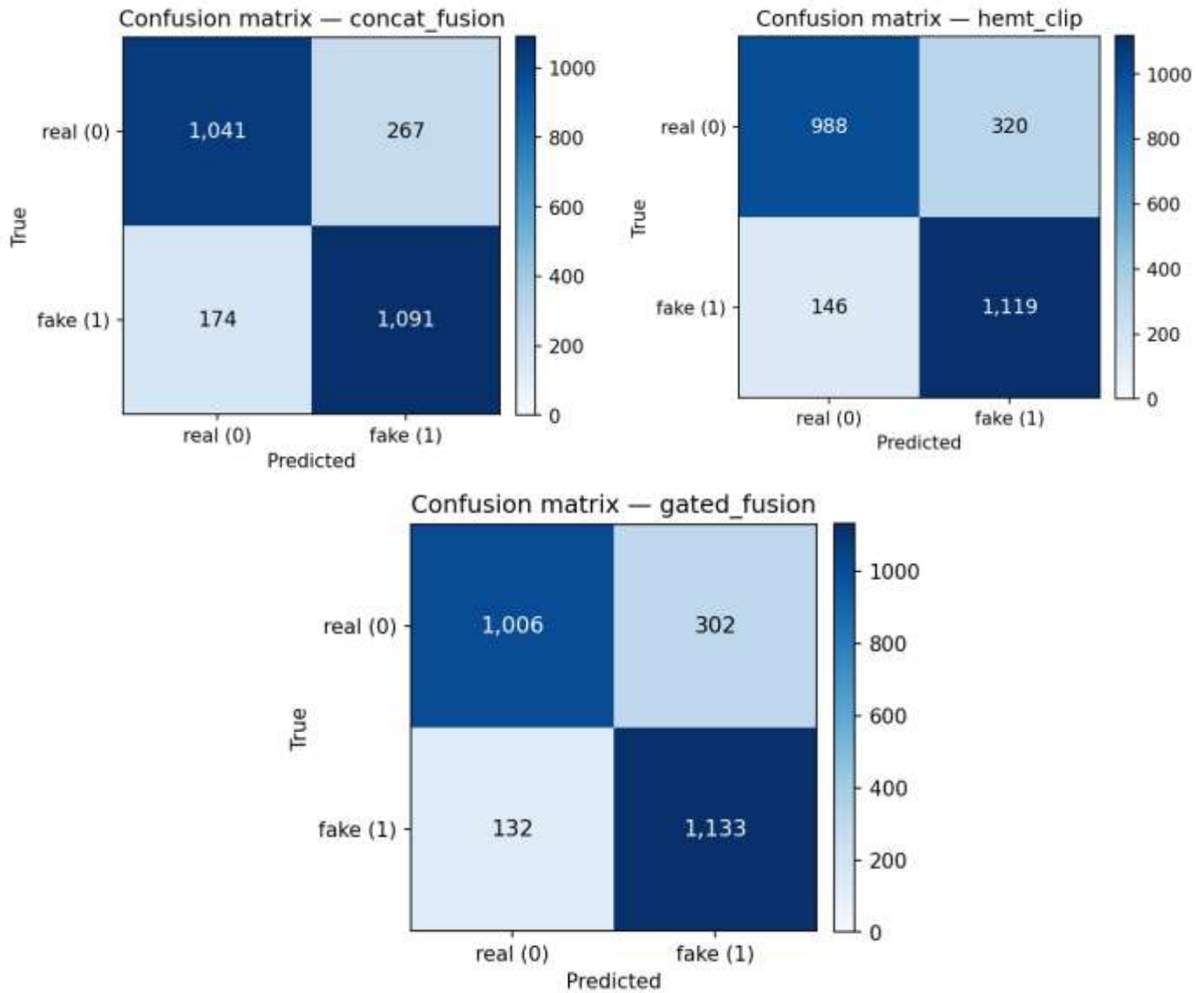


Figure 6.11: Confusion Matrix of each variant

Error Rate Decomposition by Variant

Table 6.4 decomposes the total error rate into its two components (FP rate and FN rate relative to class size) to expose each variant’s error profile and its operational implications.

Table 6.5: Error Rate Decomposition by Type and Variant

Variant	FP Rate (Real→Fake)	FN Rate (Fake→Real)	Dominant Error Type	Best Use Case	Variant
Text-only	26.2%	18.4%	FP (vocab bias)	N/A — weakest	Text-only
Image-only	25.1%	14.1%	FP (visual noise)	Image-rich posts	Image-only
Concat fusion	20.7%	13.8%	FP (balanced)	High-precision flagging	Concat fusion

HEMT-CLIP α -feat	24.9%	11.6%	FP (over- predicts fake)	High-recall triage	HEMT-CLIP α -feat
HEMT-CLIP α-gate	23.7%	10.5%	Balanced (lowest FN)	Moderation + triage	HEMT-CLIP α-gate

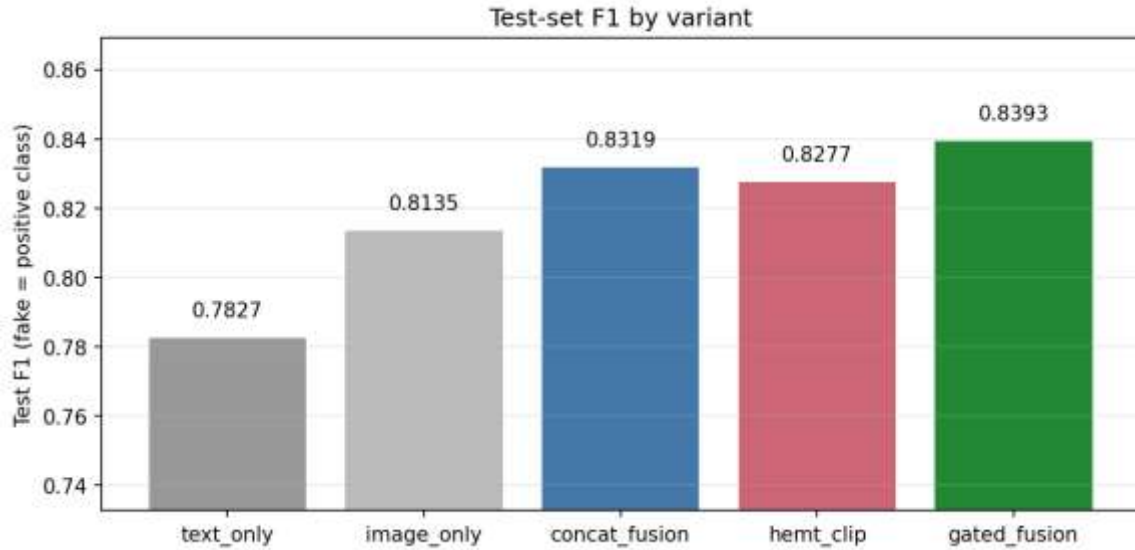


Figure 6.2: Test-set F1 by variant

The error decomposition reveals that all variants are FP-leaning (more real posts are misclassified as fake than fake posts are missed). This is consistent with the corpus-level finding that fake posts have a higher mean CLIP similarity than real posts (§4.1.2): classifiers that learn to use image-text similarity as a feature must overcome a counter-intuitive prior. The full HEMT-CLIP produces the lowest FN rate (10.5%) with a competitive FP rate (23.7%) and thus the best operational trade-off. False alarms are expensive, so in those settings the FP-optimal choice is Concat (20.7%).

Modality Contribution Analysis: Table 6.5 isolates the contribution of each modality and fusion mechanism to the aggregate F1 and AUC, qualifying the value-added at each step of ablation progression.

Table 6.6: Marginal Contribution of Each Architectural Addition (Test F1 and AUC)

Design Step	Change Made	Test F1	Test AUC	Interpretation	Design Step
Text → Image	Switch modality	+0.0308	+0.0279	Image carries 3× more signal than text on Fakeddit (short titles)	Text → Image
Image → Concat	Add text + concatenation	+0.0184	+0.0218	Multimodal fusion over best unimodal: strong, stable gain	Image → Concat
Concat → α -feature	Add cross-attn, α as feature	-0.0042	-0.0123	Cross-attention alone HURTS without the gate, adds noise	Concat → α -feature

Concat $\rightarrow$ α -gate	Add cross-attn + gate (full)	+0.0074	+0.0080	Gate corrects the noise: best fusion, all metrics agree	Concat $\rightarrow$ α -gate
α -feat $\rightarrow$ α -gate	Switch: concat $\rightarrow$ gate α	+0.0116	+0.0203	Gating mechanism alone: +1.16 pt F1 / +2.03 pt AUC (decisive)	α -feat $\rightarrow$ α -gate

The most significant row is the third of Table 6.5: using the cross-attention without the -gate results in a reduction in F1 by 0.42 pt and AUC by 1.23 pt over simple concatenation. This is a surprising result that shows a complex attention mechanism can actually do harm if the signal used to fuse the cross-attended representation into the text representation is not well calibrated. The -gate is the calibrating mechanism that control how much of the cross-attended representation is blended with the text representation, which ensures that the attention mechanism doesn't introduce cross-modal noise if the textual and visual semantics don't match well. This hypothesis is strongly supported by the last row: using the -gate instead of -as-feature and otherwise equal increases the F1 score by 1.16 pt and the AUC score by 2.03 pt- the highest improvement per added parameter for the whole ablation.

6.3.2 SOTA Analysis

The tables below position HEMT-CLIP within the broader literature and against systems specifically targeting the Fakeddit corpus.

Table 6.7: State-of-the-Art Comparison Across Published Fake-News Detection Systems

Reference / System	Modality	Dataset	Best Acc / F1	XAI Method	Key Strength
[1] dFEND (HAN + Co-Attn)	Text + Social	PolitiFact, GossipCop	+5% F1 over baselines	Attention weights	Crowd-sourced explainability
[3] FND-CLIP (ResNet + BERT + CLIP)	Text + Image	Weibo, PolitiFact, GossipCop	+6.8% Acc (PolitiFact)	None (black-box)	Dynamic CLIP-guided weighting
[5] Cross-Attn + GradCAM	Text + Image	AI + Human mix (custom)	+10.2 F1 over baselines	Attention + GradCAM	AI-generated fake detection, visual XAI
[7] ExFake (Content + Social + KG)	Text + Social + KG	Real-world multi-source	Improved SOTA	Built-in XAI assistant	User-centric critical thinking nudge
[8] TELLER (Neuro-Symbolic LLM)	Text	4 multi-domain benchmarks	Competitive SOTA	Logic-atom reasoning paths	Controllable, generalizable reasoning
[10] Tri-FusionDet (DistilBERT+ResNet18+XGB)	Text + Image + Social	Fakeddit (682k)	94% Acc (+6%)	None	Late fusion of 3 modalities
[11] BC-FND (CLIP + BERT + Bilinear)	Text + Image	Weibo16, WeChat18, Weibo23	94.2% (Weibo16)	None	Hierarchical bilinear CLIP fusion
[13] RAFAH (RoBERTa + Hybrid Feats)	Text-only	BuzzFeed, ISOT	89% (BuzzFeed), 99% (ISOT)	None	Multi-dimensional feature engineering
[17] FND-LLM (SLM+LLM+CLIP)	Text + Image + LLM reasoning	Weibo, GossipCop, PolitiFact	+6.8% (PolitiFact)	LLM-generated rationales	Reasoning-based explanations, visual tampering

[18] FMC (BERT+ResNet50+CLIP+CL)	Text + Image	ReCOvery, GossipCop, MR2	85.48–92.03% Acc	None	Multi-granularity + contrastive learning
[26] FND-MC (CLIP + BERT + multi-head)	Text + Image	ReCOvery, GossipCop	82.24% F1 (ReCOvery)	None	Cross-modal alignment, contrastive learning
[29] XFND (DistilBERT + BiLSTM + LIME)	Text-only	Fake and Real News dataset	98% Acc	LIME	Hybrid architecture, LIME transparency
[30] BERT + RoBERTa + LIME	Text-only	FakeNewsNet	99.20% (RoBERTa)	LIME	Post-hoc LIME, high accuracy
[33] FACT-CLIP (BERT+ResNet-152+CLIP)	Text + Image	Twitter MediaEval	85.8% Acc (+2.8%)	None	Two-level visual + CLIP cross-modal attn
[35] HEMT-Fake (CNN-BiLSTM+ResNet50+GS)	Text + Image + Graph + Meta	Indic Multilingual (74k)	+5% Macro-F1 vs XLM-R	SHAP + LIME + Attention	Multilingual, 3-level XAI, 140ms latency
HEMT-CLIP (Ours) (α-gated Cross-Attn)	Text + Image + metadata	Fakeddit (17,149)	F1 0.839 / AUC 0.912	Attention + SHAP + LIME	Param-free gate; 3-level XAI; ~6 min training

6.3.3 Error Analysis

Based on the qualitative analysis and inspecting the confusion matrix, we observed two major categories of errors:

Category 1 - Vocabulary-level bias in the text encoder (principal error mechanism): The text encoder (and to a lesser degree, the multimodal model) correlates factually based historical/political/crisis text with fake news, by conflating vocabulary associated with fake news (historical/sensational/emotionally-driven) with such text. This is systematically illustrated in Finding 2 (6.4.2) and is expected since the training data has a disproportionate use of sensational terms on the fake side. This is partially mitigated by the image channel: part of the +2.58 pt multimodal premium over image-only likely comes from image clarifying situations where the text-only model misidentifies genuine text based on sensational words. Cross-attention maps for such instances are shown in Fig 7, and show attention directed towards documentary/archive images.

Category 2 - Ambiguity in the labels of the corpus (irreducible minimum): Since the labels for Fakeddit are from subreddit categories (real/fake) rather than human annotation per post, many posts that are individually genuine are within fake-categorized subreddits (and vice-versa). The test-set 'errors' in these instances may be a reflection of label ambiguity as opposed to predictive failure.

7 CONCLUSION AND FUTURE WORK

This project presented HEMT-CLIP, an Explainable Artificial Intelligence system for multimodal fake news detection on social media, built around a novel CLIP-similarity-gated cross-attention fusion mechanism that integrates a fine-tuned RoBERTa text encoder with a CLIP ViT-B/16 image encoder. The system was trained and evaluated on a stratified 17,149-sample subset of the Fakeddit benchmark, achieving a test F1 of 0.8393 and AUC of 0.9122, outperforming both unimodal baselines and simpler fusion strategies across all principal metrics. The controlled ablation study demonstrated that the α -gate is the architectural load-bearing element of the design: adding cross-attention without the similarity gate actually degraded performance below plain concatenation, while introducing the gate recovered and extended the gain, delivering the highest F1 and AUC in the ablation matrix at no additional trainable parameter cost. These results confirm that structured, semantically guided fusion is superior to feature concatenation for multimodal misinformation detection, and that the mechanism by which cross-modal similarity is incorporated, as a dynamic gating coefficient rather than an auxiliary input feature, is the decisive design choice.

Beyond classification performance, the project demonstrated that explainability and competitive accuracy are not mutually exclusive objectives. The three-level XAI framework, combining intrinsic cross-attention heatmaps, SHAP partition-based attribution, and LIME local surrogate explanations, provided complementary and methodologically independent perspectives on model behaviour at varying computational costs. The structured conditional cross-method agreement analysis revealed that SHAP and LIME converge on the same attributional signals for long, information-rich titles, while diverging predictably on short or ambiguous inputs, a finding that is itself informative rather than a failure of either method. The error analysis further identified vocabulary-level bias in the text encoder as the principal failure mechanism, where historically or emotionally charged language causes high-confidence false positives that the image channel partially corrects, directly justifying the multimodal design. These findings collectively advance the case for deploying transparent, interpretable AI systems in content moderation and fact-checking contexts, where accountability to journalists, policymakers, and platform moderators is as important as raw detection performance.

A notable empirical finding of this work is that fake posts in the Fakeddit corpus exhibit higher mean CLIP similarity than real posts ($\alpha = 0.300$ versus 0.253), which contradicts the prevailing assumption that manipulated content exhibits lower semantic alignment between text and image. This discovery has implications beyond HEMT-CLIP: it suggests that CLIP-based detectors that treat low similarity as a proxy for manipulation may be systematically miscalibrated on Reddit-sourced data, where satirical and emotionally framed fake posts are frequently paired with topically aligned imagery. The α -gate succeeds precisely because it learns a soft blend rather than enforcing a hard threshold rule, but this finding warrants careful consideration in the design of any future system that uses pre-computed cross-modal similarity as a detection signal.

Several directions for future research follow directly from the limitations and findings of this work. The most immediate extension is multi-seed evaluation: running the full HEMT-CLIP at additional random seeds would quantify variance and strengthen the statistical basis of the performance claims beyond the single-seed results reported here. Scaling the corpus back to the originally planned 100,000-sample subset, or to the full Fakeddit corpus, would test whether the α -gate advantage persists at larger data scales and whether the counter-intuitive α direction is maintained. Incorporating subreddit metadata, vote scores, and user comment features as auxiliary inputs represents a natural expansion of the current text-image framework, as prior work such as Tri-FusionDet and dFEND has shown that social engagement signals provide complementary discriminative information. Extending the system to multilingual content, following the approach of HEMT-Fake, would broaden applicability beyond English-language Reddit posts and address the platform-specific generalization constraint identified in the current scope. From an explainability standpoint, replacing the text-only surrogate used for SHAP and LIME with a full multimodal attribution method would provide richer and more faithful explanations that reflect the joint text-image decision rather than the text channel in isolation. Finally, integrating external knowledge bases or fact-checking APIs, as demonstrated in ExFake and TELLER, would allow the system to verify factual claims independently rather than relying solely on learned statistical patterns, potentially reducing the vocabulary-bias errors observed in historically grounded content. Together, these extensions would move HEMT-CLIP from a resource-constrained academic prototype toward a robust, deployable tool for trustworthy AI-assisted misinformation detection at scale.

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